Firms That Automate: Evidence and Theory

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Advanced technologies have the potential to boost productivity, and both create and displace jobs. Research on automation has typically looked at aggregates. This paper dives deeper into the heterogeneity among the agents making the adoption choice: firms. I compile a set of novel findings regarding firm adoption across a range of cutting-edge technologies, such as Artificial Intelligence and Cloud Computing. Leveraging the panel dataset, I investigate the relationship between technology, employment, and productivity. I find technology adopters grow faster once they start using automation technologies. I embed technology adoption in a heterogeneous firm model to investigate the general equilibrium implications of automation. The model can reconcile my firm-level evidence of technology adoption boosting firm size, with the various macro studies suggesting a negative overall effect on employment: the general equilibrium effect on prices, entry and exit are crucial.
JEL: D22, J23, J24, L11, O14

"We've been seeing specialized AI in every aspect of our lives [...] and it promises to create a vastly more productive and efficient economy"

– Barack Obama

I. Introduction

Economists have long been interested in the impact of new technologies on productivity (Bartelsman, Van Leeuwen and Nieuwenhuijsen, 1998; Kwon and Stoneman, 1995), labour markets (Graetz and Michaels, 2015; Acemoglu and Restrepo, 2020) and growth (Solow, 1956; Romer, 1990). A new wave of research has considered the impact of automation technologies, which can perform tasks that - until recently - were limited to the domain of human labour. The sharp increase in the capability and spread of automation technologies has been widely discussed and studied (Acemoglu and Restrepo, 2016; Susskind, 2017). However, most of the empirical analysis has taken place at a macro level. For example, studies of how national robot adoption has impacted labour markets are quite commonplace (Graetz and Michaels, 2015; Acemoglu and Restrepo, 2020; Dauth et al., 2021; Kariel, 2021).

This paper fills a noticeable gap in the literature: to study the adoption of automation technologies across firms. This research has both empirical and theoretical components. Firstly, I use a novel panel dataset of Italian firms to look at the heterogeneity in adoption, and the impact, of new automation technologies. This data has five main advantages over

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many existing studies on automation among firms (Kwon and Stoneman, 1995; Bartel, Ichniowski and Shaw, 2007; Dinlersoz and Wolf, 2018; Kromann and Sorensen, 2019; Zator, 2019): (1) it includes a wide range of cutting-edge automation technologies, (2) there is information on *when* firms begin automating, (3) the data is from recent years, (4) the panel component of the data, and (5) there is a relatively large sample of firms. So, what do I find? The static results suggest automating firms are larger, pay higher wages, and are more productive. Looking over time, I find that these firms have higher growth rates. Finally, I find that firms grow faster once they become automaters.

Furthermore, I want to highlight the general equilibrium effects of automation with heterogeneous firms. I want to understand what automation among incumbents means for competition, aggregate productivity, the rate at which startups enter and at which firms shut down. This is achieved with a model of automation at the firm level, informed by the results from the empirical study. The calibrated model allows me to reconcile the evidence that automation technologies displace jobs (Acemoglu and Restrepo, 2020), with my firm-level results that it boosts firm growth. It turns out that the general equilibrium effect is crucial: automating firms expand, prices fall, and low-productivity firms exit.

As highlighted in Seamans and Raj (2018), economists "lack an understanding about how and when robotics and AI contribute to firm-level productivity, the conditions under which robotics and AI complement or substitute for labor, how these technologies affect new firm formation, and how they shape regional economies." This has been primarily due to a lack of available data on technological adoption at the firm level.

A number of recent studies have attempted to respond to this problem, utilising firm-level automation data from the USA (Dinlersoz and Wolf, 2018), Denmark (Kromann and Sorensen, 2019; Humlum, 2019), Germany (Zator, 2019), China (Cheng et al., 2019) and Spain (Koch, Manuylov and Smolka, 2019). However each study has its drawbacks, in terms of the range of time of analysis, variety of automation technologies, or small sample of firms. Furthermore, only the Koch, Manuylov and Smolka (2019) paper looks at the relationship between automation and productivity.

This nascent field of research paints a picture of various stylised facts about firms that automate. They tend to be **larger** (Zator, 2019; Bartelsman, Van Leeuwen and Nieuwenhuijsen, 1998; Koch, Manuylov and Smolka, 2019; Cheng et al., 2019; Acemoglu, Lelarge and Restrepo, 2020), have a **higher capital share** (Dinlersoz and Wolf, 2018; Bartelsman, Van Leeuwen and Nieuwenhuijsen, 1998; Cheng et al., 2019), pay **higher wages** (Dinlersoz and Wolf, 2018; Zator, 2019), are more likely to **export** (Zator, 2019; Bartelsman, Van Leeuwen and Nieuwenhuijsen, 1998), and are more **productive** (Dinlersoz and Wolf, 2018; Kromann and Sorensen, 2019; Zator, 2019; Bartelsman, Van Leeuwen and Nieuwenhuijsen, 1998; Kwon and Stoneman, 1995; Koch, Manuylov and Smolka, 2019).

I find similar results, and can go into more detail thanks to a rich dataset on Italian firms. I use the Bank of Italy's "Survey of Industrial and Service Firms" firm-level automation data to investigate the relationship between automation, productivity, firm dynamics and employment. This survey has been conducted over a long time, for a large sample of firms, asking a wide range of automation-related questions, including recent questions on "Industry 4.0". For example, it asks about the use of Artificial Intelligence (AI), the

Internet of Things (IoT), and Industrial Robotics. This survey has approximately 5,000 firms in the panel, and covers 1984 - 2019. It is an excellent dataset to examine which firms automate, how this has changed over time, and the impact of these choices. The first stage of the research will be to analyse the distribution of automation across firms, investigate variation between industries, and provide descriptive statistics on firm-level automation. I also leverage the panel dataset to better understand how automation affects firms.

I find that technology adopters grow faster than non-adopters, but there is little evidence of this difference for blue-collar employment or productivity. By focusing on the impact *on adoption*, I continue to find significant overall employment effects, suggesting the contribution of selection effects is muted. Again, there seems to be a limited impact on unskilled work and productivity (apart for firms using robotics). In order to fully get an idea of what happens to firms on adoption of technologies, I conduct event studies which show considerable employment effects from automation technologies, in the range of 4 - 11% 5 years after adoption.

A model of automation is built, extending the Hopenhayn (1992) framework. This is a workhorse model of industry dynamics which lends itself to questions about the size distribution of firms, entry and exit behaviours and the impact of policies (e.g. a robot tax). Automation is built into the production function through a task-based framework (Acemoglu and Restrepo, 2016), where such technology can displace routine workers, but complements non-routine labour. The calibrated model produces predictions about the effect of automation, and can then be compared to the data.

The intuition behind the model is simple: automation technology is assumed to be cheaper than human labour, but it requires some fixed set-up and implementation costs. Thus only the most-productive and profitable firms can automate, because they can afford the fixed costs. So the effect of automation may be partly due to selection, as well as the reallocation of output towards already-productive firms, rather than due to it being a huge efficiency-saving technology.

The model can be used to look at how automation affects aggregate productivity, through selection, reallocation and entry/exit. It can also be used to look at the impact on wage distributions, and on the life-cycle dimensions of firm distributions.

There are three important questions regarding automation and productivity. Firstly, does automation drive productivity growth in firms, or is it simply that only productive firms can automate (due to costs or risks involved)? Causal inference methods will be important here, along with approaches to control for selection. Secondly, I can decompose industry productivity growth into the direct effects from automation, and the indirect effects from reallocation of output towards automating firms (Koch, Manuylov and Smolka, 2019). Finally, it may be that the general equilibrium effects of automation on wages and prices induces more entry of less-productive firms, which leads to potentially ambiguous effects on aggregate productivity. The model and data will be used in tandem to evaluate this question.

The paper is organised as follows. Section II presents a brief literature review, covering the economics of automation, existing research on the various technologies in this study, and heterogeneous firm models. In Section III I describe the firms that adopt the various

technologies, and perform a number of analyses to better understand the relationship between automation and employment. The model is described in Section IV, along with the calibration and results.

II. Literature Review

This research touches upon three strands of literature: the economics of automation, writ-large; the analysis of the adoption and impact of specific technologies; and theoretical models of heterogeneous firms.

Automation

The fast-growing literature on the economics of automation has a variety of branches, considering different countries, technologies, and theoretical frameworks. New technologies which automate tasks can affect the labour market in a variety of ways. Firstly, workers are most likely to benefit if they produce labour tasks which are complemented by automation technologies (Autor, 2015). This is somewhat related to skill-biased technical change (SBTC), as those with higher skills - and perhaps skills more complementary to machines - have faced the greatest rise in demand (Griliches, 1969). Secondly, wage gains depend on the elasticity of labour supply; if these complementary skills are abundant in the labour market and workers have elastic supply, workers may flow into the automated industry, somewhat mitigating the income growth (Autor, 2015). Thirdly, output elasticity of demand and income elasticity both play a crucial role. Autor (2015) highlights the impact of the former by noting that the share of expenditure on food has fallen following productivity improvements in agriculture, while the share on healthcare has risen with gains in productivity. The importance of income elasticity is most clearly related to Baumol's 1967 distinction between "technologically lagging" sectors and those at the vanguard of automation. Rising productivity in leading sectors may boost employment in the laggards, where demand for the final goods are strongly income elastic. The spillover effects of productivity improvements essentially arise from a rise in output where larger untapped demand exists, and the effects have been found to be consistently positive (Autor and Salomons, 2017).

An important strand of research has highlighted that jobs are not "lumps of labour" (Susskind and Susskind, 2015) but a combination of tasks which may require distinct skills and inputs. Many recent economic models of automation embed this concept in the production function, by assuming work consists of a continuum of tasks, increasing in cognitive difficulty. For a given level of technological progress, machines can replace labour up to a certain point, but beyond that we require human input. For example, this may include tasks for which face-to-face contact is still essential, or the variety and complexity of tasks has not yet been competently replicated by AI, such as private hire driving, managing other people, or therapy. This task-based framework allows us to identify different effects of automation: the "displacement effect" whereby workers are replaced, putting downwards pressure on wages and employment; the "productivity effect" which raises demand for labour in non-automated tasks; "capital accumulation" if automation raises demand for capital, and thus for labour; "capital deepening" which is simply improved productivity in already-automated tasks (Acemoglu and Restrepo, 2018). Therefore, the overall impact of adopting automation technology is ambiguous. Furthermore, it is likely to depend on the complementarity between the technologies and various labour skills, along with the

varying demand for different tasks. Finally, the impact will depend on the creation of new tasks, in which humans still have a comparative advantage. In the task-based framework, this can be modelled as the continuum extending, which under certain assumptions leads to a stable balanced growth path (Acemoglu and Restrepo, 2016), rather than the eventual destruction of all jobs (Susskind, 2017).

The empirical research has used a variety of datasets and identification methods to consider the relationship between labour markets and automation. Many studies using national industrial robotics data have found a relationship between this technology and employment, but the effect varies across countries, from positive (Graetz and Michaels, 2015), to negative (Acemoglu and Restrepo, 2020), to sector-specific (Dauth et al., 2021). Clearly, the validity of these results rests on the instruments successfully eliminating endogeneity issues, such as cost-push shocks which simultaneously increase the incentive to adopt robots and reduce demand for labour. Furthermore the use of national robots data, rather than a more disaggregated measure, may yield misleading inference, especially if there is significant heterogeneity.

Thus the recent firm-level research may provide more insight. Bessen et al. (2019) take advantage of Dutch micro-data to show that increased expenditure on automation increases the probability of worker separation and reduces employment at the intensive margin, with no impact on wage rates. On the flip side, Dechezleprêtre et al. (2020) consider the opposite direction of causality, and they show their novel measure of firm-level automation innovations rises with exogenous shocks to low-skill wages. This evidence suggests firms respond to changes in factor prices by making more investments in innovations to replace low-skill workers.

Recent studies on the impact and heterogeneity of automation by firms covers a range of countries and technologies. For example, Acemoglu, Lelarge and Restrepo (2020) investigate robot adoption by French manufacturing firms, with adopters boosting their employment at the expense of rivals, but also experiencing declines in labour share. Likewise, Koch, Manuylov and Smolka (2019) examine robot adoption by Spanish firms, and find similar gains in output, employment, and a falling labour share, as well as reallocation towards adopters. Zator (2019) shows that adoption of digitization and automation technologies is associated with local labour scarcity, although variation across technologies and industries is important. Notably, they find similar substitution effects with robots as in Acemoglu, Lelarge and Restrepo (2020), but complementarities between digitization and labour in some service industries.

Advanced Technologies

The technologies discussed in the empirical part of this paper are Mobile & Cloud Computing, Artificial Intelligence & Big Data, Internet of Things, Industrial Robotics, and 3D Printing. These cutting-edge technologies have the potential to automate a wide range of tasks, and are likely to have a significant impact on firm productivity, workers and competition. There is a nascent but growing literature analysing these technologies. I will go through each in this section.

Mobile & Cloud Computing

Cloud Computing involves using remote servers to manage data, and can encompass a range of technologies such as software (e.g. Google Docs), platforms (e.g. Microsoft Azure), and data storage (e.g. Amazon S3) (Bayrak, Conley and Wilkie, 2011). There is very little economic research on the adoption and impact of Cloud Computing. Bayrak, Conley and Wilkie (2011) highlight this gap in the literature, and they underscore that Cloud Computing doesn't necessarily permit the completion of tasks that were unavailable with previous technologies. However, it does reduce fixed costs, and offers faster scalability than before. This insight is theoretically grounded in Etro (2009) in a DSGE framework: they conclude that Cloud Computing can raise the growth rate and employment through the entry of new small- and medium-sized enterprises (SMEs). In fact, Etro (2009) estimates that the largest medium-run impact in the EU would occur in Italy, with approximately 80,000 new jobs under fast adoption.

Artificial Intelligence & Big Data

Artificial Intelligence (AI) is "the capability of a machine to imitate intelligent human behavior" (Merriam Webster, 2020) and has been subject to a lot of discussion within the economics profession. It is sometimes used interchangeably with 'automation'. For example, Aghion, Jones and Jones (2017) study the impact of AI on long-run growth, extending the Zeira (1998) model (which itself considers 'machines' that are more akin to robots) and they regularly employ the language of tasks being 'automated' or not. However, AI may not *always* simply be an automation technology, and may need to be combined with robots, big data, cloud computing, or other technologies, in order to automate a task. It seems more reasonable to think of the common tasks of AI: decision-making, prediction, anomaly detection, among others (Babina et al., 2020). In fact, it is these more specific applications that are asked about in the survey I use for this paper. There are both theoretical (Aghion, Jones and Jones, 2017; Acemoglu and Restrepo, 2018) and empirical (Babina et al., 2020; Zator, 2019) contributions on the economic impact of AI: however, as yet, there is limited *direct* evidence on firm adoption of this technology, due to a lack of data.

Big Data is the availability of real time, larger scale, more varied, unstructured data (Einav and Levin, 2014). It is often reasonable to combine AI with Big Data as the former typically depends on the latter. In this survey, AI and Big Data are combined when firms are asked in 2015, but separated in the 2017 wave. It is more common to see economists discuss the opportunities of Big Data as a research tool (Einav and Levin, 2014), rather than the impact it will have on firms.

Internet of Things

The Internet of Things (IoT) is a set of technologies which connect machines and the internet. They are part of an infrastructure that connects sensors and actuators to computer systems via networks, and these systems can track the performance and actions of

machines through the internet (Manyika et al., 2015). There is limited economic research on the impact of these technologies, although Edquist, Goodridge and Haskel (2019) find significant relationships between TFP growth and the number of IoT connections per person. This country-level analysis likely suffers from endogeneity concerns, as the authors note, so can really only be interpreted in terms of high-level correlations. The firm-level survey - and panel component - allows me to investigate IoT with finer detail (across industries, firm age etc.) and hopefully alleviate concerns of endogeneity.

Industrial Robotics

Research on the adoption and impact of industrial robots is widespread relative to the other technologies in this paper. The findings are mixed, with evidence of displacement of jobs in the U.S., while there was no overall fall in employment in Germany, but job composition was affected (Acemoglu and Restrepo, 2020; Dauth et al., 2021). Similarly, there is evidence robot adoption led to changes in the skill composition of labour across Europe, alongside increases in labour productivity (Graetz and Michaels, 2015). However, these studies use quite aggregated robots data, which may mask more interesting behaviour occurring where the robots are purchased: the firms.

More recently, there has been firm-level research on the impact of robots (Acemoglu, Lelarge and Restrepo, 2020; Koch, Manuylov and Smolka, 2019; Cheng et al., 2019; Stapleton and Webb, 2020). There are some consistent findings, for example that robot-adopting firms tend to be larger, and that adopters tend to grow, while their non-adopting rivals shrink (Acemoglu, Lelarge and Restrepo, 2020; Koch, Manuylov and Smolka, 2019).

3D Printing

3D printing technology uses a digital depiction of an item alongside "additive manufacturing" techniques to create physical objects (Bechtold, 2015). There is little existing research on the economic impact of this technology, and the findings are mostly descriptive, context-specific, or lacking a theoretical and empirical basis (Weller, Kleer and Piller, 2015; Bechtold, 2015; Reeves and Mendis, 2015). One exception is work by Abeliansky, Martínez-Zarzoso and Prettner (2015) which introduces 3D printing in a Melitz (2003) framework such that countries facing high trade costs will be faster adopters of this technology. Their findings support this theoretical prediction, but the analysis does take place at a country level.

Heterogeneous Firms

My model builds on Hopenhayn (1992) and Hopenhayn and Rogerson (1993), alongside a task-based framework for the production technology (Acemoglu and Restrepo, 2016) to allow for routine and non-routine labour, and an automation technology. There is a rich literature using heterogeneous firm models to understand the relationship between entry, exit and aggregate productivity. One major benefit of using such models is the focus on the long-run effect of reallocation in response to job and firm turnover, thanks to the

stationary equilibrium solution approach. Thus, we can analyse the effect of changes in policy, technology, or other exogenous shocks.

For example, Clementi and Palazzo (2016) show that entry and exit can propagate aggregate shocks, leading to greater persistence of such shocks. The intuition is that a rise in a common productivity component induces entry of firms with low idiosyncratic productivity, offsetting the former effect somewhat. However, the long-run effect is positive due to the firm distribution shifting towards young, growing, and efficient firms. This highlights how the dynamics of net entry, and selection effects, are important mechanisms in such models.

Closer in spirit to the question in this paper, Asturias et al. (2020) analyse the effect of policy reforms on barriers to technology adoption, which leads to a boost in aggregate productivity, with an increase in the contribution of entry and exit. The change in productivity can be decomposed in various ways to identify the mechanisms at play (Petrin and Levinsohn, 2012; Foster, Haltiwanger and Krizan, 2001).

Finally, two important papers take a similar approach of combining microdata on firms and technology with a structural model of the macroeconomy. Work by De Ridder (2019) and Lashkari, Bauer and Boussard (2019) look at the impact of software and Information Technology (IT), bringing together novel empirical insights on firms (from French microdata). They each build structural models to delve into the implications for market power and productivity. The research by De Ridder (2019) presents a mechanism to explain the slowdown in productivity growth, decline in business dynamism and rise in market power, all as a function of the rise in *intangibles* such as software, which reduce marginal costs and raise fixed costs. My model also embeds the idea that automation technology affects both of these costs. Crucially, this paper speaks to the scalability of intangible inputs, and the relative efficiency of such inputs differing across firms. I differ from this assumption by introducing a Markov process for firm productivity, and the endogenous automation choice depends on this. Somewhat differently, Lashkari, Bauer and Boussard (2019) frame IT inputs as having marginal product rising with scale: it helps deal with organisational limits. Their model implies endogeneity of firm-level returns to scale, which leads to larger firms using more IT, and a lower labour share. Such results are also found in the model I build.

III. Empirical Findings

In this section I introduce the data, then describe the different measures of automation, and the patterns across firms. I begin with an overview of automation: looking at aggregates and distributions of firms that automate. I find that adopters of automation tend to be larger, but do not vary systematically by age.

Additionally, I look more closely at which firms automate across the various technologies. Again, there is evidence that automating firms are larger, but there is no relationship with age. Technology adopters have higher turnover, and have higher productivity. In particular, firms using 'digital' automation technology look quite different, on average, from those using 'physical' machinery. Then I investigate how technology adopters differ from non-adopters over time. I find that automating firms tend to grow at faster rates, and this fact holds across the size and age distribution.

I look at the patterns of 'switchers' - that is, firms that become automation adopters, or stop using automation. I find that firms both adopt and drop technologies, suggesting that investment is reversible. I use propensity scores to 'predict' future adoption, and find it about as accurate as using contemporaneous firm characteristics.

Finally, I use the propensity scores, and the panel, to try to shed some light on how these technologies affect firm outcomes. Each method I use - propensity score matching, two-way fixed-effects models, and event studies - suggests that automating firms grow faster than non-adopters. In particular, it seems that technology adoption is associated with a boost to employment, but only for skilled workers.

Data

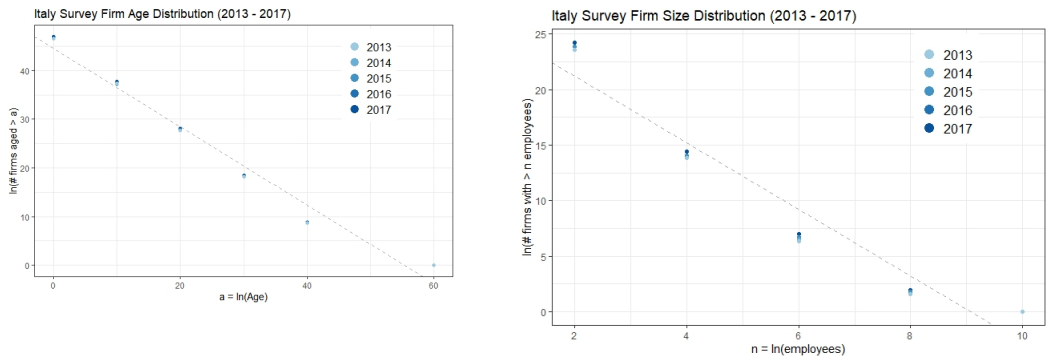
The data is from the Bank of Italy "Survey of Industrial and Service Firms". This survey asks a wide range of questions from a representative sample of around 5,000 firms with at least 20 employees. Since around 2000, the survey has included service firms and smaller businesses, to better represent the Italian firm distribution. The survey uses a one-stage stratified sample design. The strata are selected according to sector, size (average annual number of employees), and region of head office. Firm-year weights are computed with Horvitz-Thompson estimators, and post-stratification adjusted weights are computed using outside information of certain geographic characteristics. Further details on the survey design can be found in the Appendix B.

Firms were asked questions on automation technologies in 2015 and 2017. In each year, firms are asked about the use of distinct technologies, which have the potential to replace labour inputs in certain tasks. Firms give a binary response on the current use of each technology, alongside the number of years since adoption. In 2015, the technologies are: Mobile Broadband and the Cloud; Artificial Intelligence (AI) and Big Data; the Internet of Things (IoT); Industrial Robotics using AI; 3D printing. In 2017, firms are also asked about E-Commerce, and separately about Cloud Computing, AI and Big Data. Furthermore, in both 2016 and 2017, firms are asked the share of total investment which was spent on advanced digital technologies.

Italian Firms

Firstly I will introduce some general facts about the firms in this survey. Firm age and size distributions seem to follow an approximately Pareto distribution, as has been found elsewhere in the literature (Axtell, 2001; Garicano, Lelarge and Van Reenen, 2016; Segarra and Teruel Carrizosa, 2012).

Figure 1. : Firm Distributions



The average firm is 36 years old, has 50 employees, and earns over €10.7 million in turnover. The average firm wage per worker is around €29,600.

Automation: An Overview

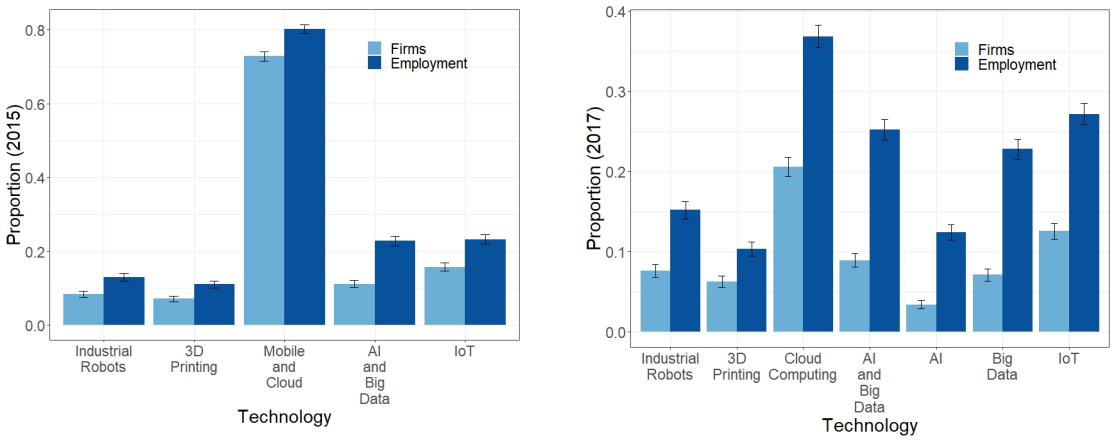
In 2015, the questions were answered by approximately 3,400 firms. This rose to around 3,800 firms in 2017. The most-used advanced technologies are Mobile & Cloud Computing (73% - although we can infer this is mostly the use of Mobile, as adoption of Cloud in 2017 is just 21%), the Internet of Things (13 - 16%) and E-Commerce (22%). The least-used are 3D Printing (6 - 7%), Industrial Robots (8%), AI and Big Data (9 - 11%).[1]

It is instructive to compare the proportion of firms using each technology, and the share of total employment in these firms. Overall, firms adopting *any* of these automation technologies take up a much larger share of employment: around 30 - 33% of firms are adopters, employing 43 - 51% of workers. The graphs in Figure 2 split this relationship by technology.

Clearly for all technologies, the firms adopting them are larger than non-adopters. This relationship is strong for both 'digital' automation technologies (Cloud, AI, Big Data, Internet of Things), and for 'physical' technologies (Robotics, 3D Printing). More specifically, around 21 - 29% of firms adopt at least one of these 'digital' automation technologies,

[1]The raw and weighted proportions of firms using these technologies is summarised in Tables F4 and F6 in Appendix F.

Figure 2. : Technology Adopters: Proportion of Firms and Employment



yet employment in these organisations is 35 - 47% of the total. Meanwhile, 10 - 11% of firms use Industrial Robots or 3D Printing, employing 18 - 20% of all workers.

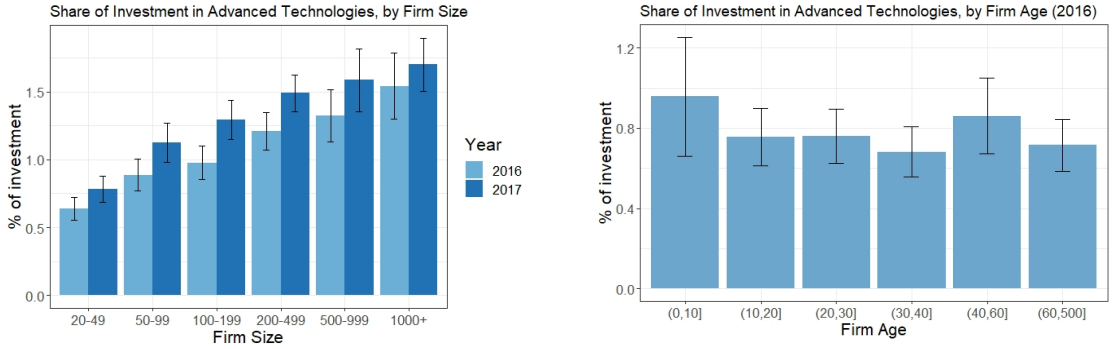
It seems plausible that firms adopting one of these advanced technologies are likely to adopt others. This bears out in the data (see Tables F8 and F9 in Appendix F), but the overlap varies across technologies. Unsurprisingly, firms adopting AI overwhelmingly use Big Data in 2017 (90%). These two technologies go hand-in-hand; the benefits of AI and Big Data depend significantly on use of the other. Likewise, there is significant adoption of both of the more industrial technologies: Robots and 3D Printing (84% and 86% in 2015 and 2017). Even for technologies which may seem less complementary, such as Cloud Computing and Industrial Robots, the proportion of firms using both is 76% in 2017.

The share of total investment allocated to advanced digital technologies increases systematically with size, but not unconditionally with age. There is also a noticeable increase between 2016 and 2017 in the investment share in advanced technology across the size distribution. These findings are shown in Figures 3.

The joint age $\times$ size distribution of technology spending in 2016 also suggests that automation investment rises only with firm size. However, in 2017 there is some suggestion that a greater share of investment is jointly increasing in age and size. Plots of these distributions can be found in Figures B1 in Appendix D.

Regressing the share of investment in advanced technology on firm size and age, with sector and region fixed-effects, highlights that larger firms invest a greater proportion in these technologies. As the joint age $\times$ size distribution highlighted, it seems that age is not associated with investment shares in automation in 2016, but it is in 2017. In 2016, the estimated coefficients in Table 1 show that larger firms invest significantly more in automation, even controlling for age, sector and region, with an increase of 0.25% for

Figure 3. : Investment Share on Advanced Technology Varies with Firm Size, but not Age



each percentage rise in employment.[2] In 2017, we can see that the relationship between technology investment shares and employment has risen slightly to around 0.3%, while the estimated coefficients on age are positive and significant.

Table 1—: Estimated Coefficients from Advanced Tech. Investment Regressions

Dependent variable: *Share of Investment in Advanced Tech.*						
	2016			2017		
log(Emp.)	0.279***	0.278***	0.254***	0.337***	0.329***	0.299***
	(0.025)	(0.025)	(0.026)	(0.028)	(0.028)	(0.028)
Age		−0.0000004	0.0002		0.0034**	0.0026*
		(0.001)	(0.001)		(0.001)	(0.001)
Sector FE			✓			✓
Region FE			✓			✓
N	3756	3749	3749	3926	3926	3926

Note: Estimates are significant at levels of 0.1%: ***, 1%: **, 5%: *.

The spending on advanced technology is also unequally distributed across sectors, with the greatest share in Metal Manufacturing (1 - 1.25%), and the lowest in Hotels & Restaurants (0.3 - 0.4%). A full breakdown of spending by sector is in Figures B2 in Appendix B.

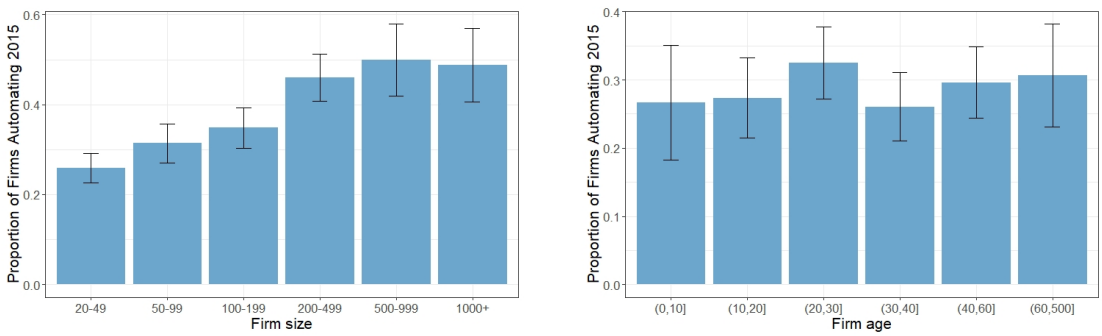
[2]The regression specification of $Y = \beta \ln X + \epsilon$ gives $\beta = \frac{dY}{\%\Delta X}$ which, in this case, is the change in the share of investment in advanced technologies with a percentage change in employment.

Adopters of Automation

Adopters of automation technologies tend to be larger, but there doesn't always seem to be systematic variation by firm age. There is significant variation in the proportion of firms using advanced technologies by region, 'macro' sector (manufacturing/extraction/services) and by more disaggregated industries. It is not necessarily the case that firms that export more are more likely to use advanced technologies. Here I briefly outline some findings on how automation adopters differ, across the various technologies, with further detail provided in Section A of the Appendix.

Generally, it seems that larger firms are more likely to use advanced technologies. Figure 4 shows the proportion of firms adopting *at least* one automation technology in 2015, by employment bin. It is clear that a greater proportion of large firms use these technologies. However, there is no obvious relationship with firm age.

Figure 4. : Adoption of Automation Technologies Rises With Size, Not Age



Simple single-year regressions of technology adoption on log employment and age highlight this relationship clearly: employment is always strongly positively associated with adoption with statistical significance. The estimated coefficients on log employment range from 0.03 - 0.08, and are always significant at the 1% level or below. Thus a percentage rise in employment raises the likelihood of adoption in the 3 - 8% range. These estimated coefficients barely change with the inclusion of firm age, which is, generally, small and insignificant. It is only for AI & Big Data (negatively) and Industrial Robotics (positively) that there is evidence age is associated with adoption of technology.

In addition, I often find that more hirings increases the probability of adoption, while more firings does the opposite. This suggests that it is not only the size of the firm that is associated with technological adoption, but whether it is growing, or not. This will be investigated in more detail in the next section. The final consistent finding across all technologies is that adoption is positively related to turnover. The causation could go either way here: higher turnover may allow for investment in advanced technology, or automation may boost productivity. More specifically, the likelihood of technology

Table 2—: Estimated Coefficients from Technology Adoption Regressions 2015

	Dependent variable: *Technology Adoption*									
	Mobile & Cloud		*AI & Big Data*		*Internet of Things*		*Industrial Robotics*		*3D Printing*	
log(Emp.)	0.033**	0.034**	0.039***	0.041***	0.052***	0.052***	0.040***	0.038***	0.027***	0.025***
	(0.010)	(0.011)	(0.009)	(0.009)	(0.009)	(0.009)	(0.006)	(0.006)	(0.006)	(0.006)
Age		−0.0005		−0.0007		−0.0002		0.0009**		0.0004
		(0.0005)		(0.0004)		(0.0004)		(0.0003)		(0.0003)

Note: Estimates are significant at levels of 0.1%: ***, 1%: **, 5% *.

Table 3—: Estimated Coefficients from Technology Adoption Regressions 2017

	Dependent variable: *Technology Adoption*									
	Cloud Computing		*AI & Big Data*		*Internet of Things*		*Industrial Robotics*		*3D Printing*	
log(Emp.)	0.074**	0.073**	0.064***	0.065***	0.076***	0.076***	0.036***	0.036***	0.033***	0.033***
	(0.010)	(0.010)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.004)	(0.004)
Age		0.0005		−0.0005*		−0.00007		0.0004**		−0.00006
		(0.0005)		(0.0004)		(0.0004)		(0.0003)		(0.0003)

Note: Estimates are significant at levels of 0.1%: ***, 1%: **, 5% *.

adoption increases around 0.01 - 0.3% with each additional worker, and between 0.03 - 0.40% with a percentage rise in turnover (see Table A3 in Appendix A).

Technology adopters have significantly higher employment and wages. For example, firms using AI & Big Data employed almost 63 workers on average in 2015, compared to just under 49 for non-adopters. In addition, such firms renumerated workers with over €31,000, compared to under €29,000 for businesses not using this technology.

There is an interesting distinction regarding firm age between the 'digital' and 'physical' automation technologies: firms adopting the former are significantly younger than non-adopters, but this relationship is reversed for the latter. For example, adopters of AI & Big Data are aged 33 on average in 2015, compared to over 35 for those that don't adopt. On the flip side, the average age of robot adopters is over 41, with non-adopters close to 35. Some illustrative differences between 'digital' and 'physical' automating firms and non-adopters are contained in Table 4.

The relationship between productivity and technology use is an important question to examine. Both the correlation and the direction of causality can be informative about firm behaviour. I consider turnover per worker as a simple proxy for productivity, and regress this on technology adoption, alongside firm age and fixed-effects for sector and region. The estimated coefficients (in Table B1 in Appendix B) from this regression suggest that adoption of certain advanced technologies is associated with 9 - 17% higher in turnover per worker, and the evidence is significant for Cloud Computing, AI & Big Data, and Internet of Things - the 'digital' automation technologies. The results are not consistent with either robotics or 3D Printing being correlated with higher productivity. However, it should be

Table 4—: Automating Firms vs Traditional Firms: Summary Statistics

	Employment	*log(Hours)*	*log(Wage)*	*Age*
'Digital' Automating Firms	59.6	7.41	10.32	34.1
Non-Adopters	47.6	7.40	10.27	35.4
'Physical' Automating Firms	63.8	7.40	10.31	38.7
Non-Adopters	48.1	7.40	10.28	34.6

Note: 'Digital' automation technologies are Mobile & Cloud, AI & Big Data, and the Internet of Things. 'Physical' automation technologies are Industrial Robotics, and 3D Printing. Note that hours are average hours per worker, over the year.

noted that this measure of productivity contains employment on the denominator, so if technology adoption is associated with firm size, this could mechanically hide any potential relationship between technology and turnover.[3]

Further details on the heterogeneity in adoption across the various automation technologies can be found in Appendix G.

Finally, I leverage the panel to label firms as technology adopters, and non-adopters, from 2010 - 2018. The difference in growth rates for adopters and non-adopters is compared across firm size and age distributions. I find that average growth rates of automating firms are generally higher than that of non-adopters. This holds for most automation technologies, across the age distribution, as seen in Figure 5. The faster rate of employment growth of technology adopters is also consistent for small- and medium-sized firms, but not large firms.

Looking into how growth rates differ across the age *and* size distribution underscores the pattern of larger automating firms growing more slowly, even conditional on age (see Figure B5 in Appendix D). Therefore just based on averages over this period, it seems that technology adopters grow faster than non-adopters, unless they are already very large.

Automation Adopters Over Time

One particularly stark finding is that a non-negligible proportion of firms that use advanced technologies in 2015 did not use them in 2017. As can be seen in Figure 6, similar shares of firms use them in just 2015, or just 2017, or in both. This evidence suggests that switching to using advanced technologies isn't irreversible, and firms may respond to idiosyncratic shocks or changes in relative factor prices when making investments in such technologies.

I consider if patterns of technology adoption in 2015 are informative about future adoption. I apply propensity-score matching (PSM) on 2015 adopters, and use these scores to 'predict' adoption in 2017. The logit for PSM contains a host of potentially useful

[3]Regressing turnover on technology adoption, with a control for firm size, would do no better, as this would lead to issues of multicollinearity.

Figure 5. : Average Growth Rates (2010 - 2018) for Automation Adopters compared to Non-Adopters

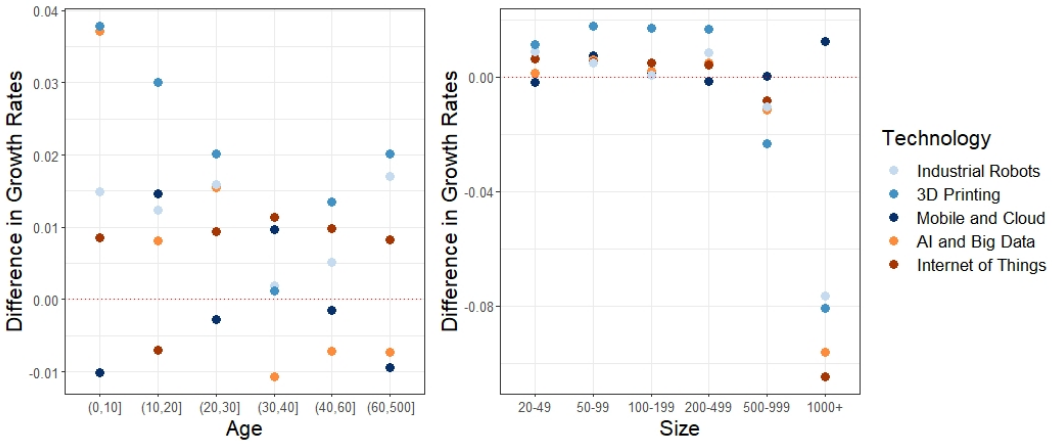
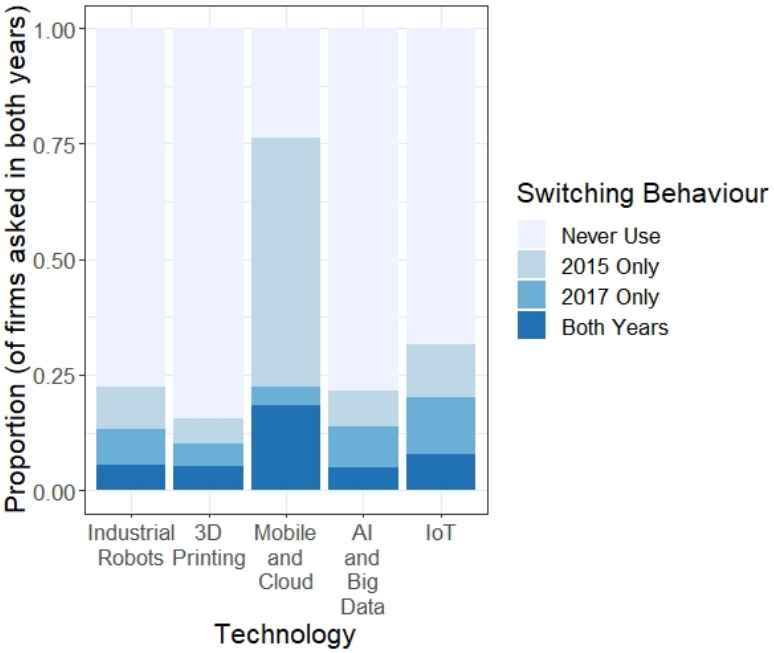


Figure 6. : Change in Use of Technologies between 2015 and 2017



explanatory variables. Further details on these, as well as the fit of the propensity score for explaining technology adoption in 2015, are contained in Section G of the Appendix.

The propensity scores from 2015 are used to predict adoption in 2017 for all firms in both waves of the survey.

I run three logit regressions for 2017 adoption. Firstly, I simply regress 2017 adoption on the 2015 propensity score. Then I consider propensity scores above or below the median as 'predicting' adoption or no adoption, respectively. The benefit of this measure it is more robust to outlying propensity scores. Finally, I run a logit on a host of 2017 covariates (the same as in the PSM specification) to explain 2017 adoption. I then compare various measures of fit to understand if there is information in 2015 adoption behaviour that can help predict adoption in later years, compared to the characteristics of firms in 2017.

It seems that the 2015 propensity scores do reasonably well at predicting future adoption of automation technology. I find that the propensity scores predict 2017 technology adoption behaviour with 62 - 64% accuracy (i.e. the proportion of firms correctly predicted as adopters/non-adopters). Another measure of accuracy - the F1 Score - also gives results in this range.[4] For comparison, when I use a wide set of 2017 variables (size, turnover, exporting behaviour etc.) to predict 2017 technology adoption, the prediction is barely any better: the accuracy is 62%, and the F1 Score is 65%. The full set of results, across all technologies, are in Table G4 in the Appendix.

Another informative result of the PSM exercise is I can used the 'matched' set of firms to compare automation adopters to 'similar' non-adopters. I regress the log of employment on firm technology adoption for this matched set of firms. Although the sample size is on the smaller side, this provides further evidence of the relationship between automation adoption and employment. The matched regression results suggest technology adopters are 33 - 82% larger than matched non-adopters, with statistical significance across all technologies.

Table 5—: Propensity Score Matching Regression Results, 2015

Dependent variable: *Log Employment*						
	Any Tech.	*Cloud*	*AI & Big Data*	*IoT*	*Industrial Robotics*	*3D Printing*
Tech. Adoption (nearest)	0.461***	0.822***	0.623***	0.475***	0.370***	0.330**
	(0.06)	(0.07)	(0.11)	(0.08)	(0.10)	(0.11)
N	1914	1376	674	1042	720	524
Tech. Adoption (full)	0.586***	0.400***	0.818***	0.583***	0.535***	0.537**
	(0.05)	(0.06)	(0.07)	(0.06)	(0.07)	(0.08)
N	2554	2580	2547	2541	2544	2538

Note: Estimates are significant at levels of 0.1%: ***, 1%: **, 5% *. Coefficients are for log employment regressed on (binary) technology adoption, with matching described in brackets: "nearest" performs greedy nearest neighbour matching; "full" performs optimal full matching, so treated and control observations are assigned to a class and each receives at least one a match.

I can leverage the large set of firms over the panel to understand the relationship between technology adoption and firm size. There are around 3,500 firms that can be labelled as

[4] As there are significantly more non-adopters than adopters, the F1 Score is preferred, as it aims to better balance the success of a classification algorithm with an unbalanced sample.

adopters or non-adopters of advanced technologies. I compile a panel of these firms over the period 2010 - 2018, giving around 25,000 firm-year observations. With this panel, I can look at how technology adopters differ over time, and how firms change before and after adoption.

Given that there is potentially non-random non-response to the questions on automation adoption, I use the class non-response adjustment factor to re-weight my panel as in Dargatz and Hill (1996). For further information on the extent of non-randomness of non-responders, and the method used to re-weight the panel, refer to Section E of the Appendix.

I employ two-way fixed-effects (TWFE) models, to estimate the difference in employment and productivity between adopters and non-adopters. Furthermore, event studies can show how firm size changes before and after adoption.

In order to understand how technological adopters differ from non-adopters over the panel, I estimate a two-way fixed-effects (TWFE) model. Employment, blue-collar employment, and labour productivity (proxied by turnover per worker) are regressed on firm and time fixed-effects, along with controls for firm region, sector and age, and the variable of interest: a binary technology adoption variable (i.e. does firm i ever adopt the technology?). As treatment effects may be heterogeneous over time, I interact the binary technology adoption with the year.

$$\ln Y_{it} = \alpha_i + \gamma_t + \delta X_{it} + \beta_t \mathbb{1}\text{Tech}_i \times \text{Year}_t + \epsilon_{it} \tag{1}$$

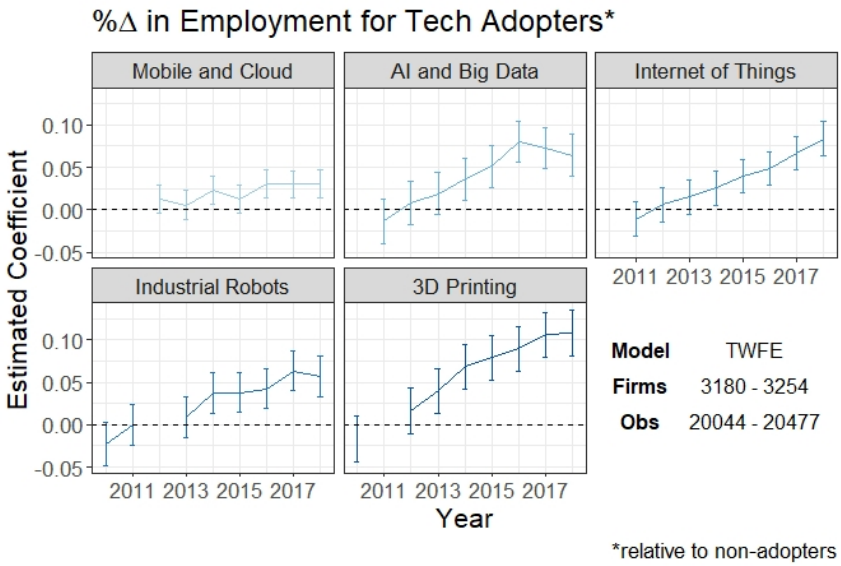
The coefficients β_t describe how technology adopters behave over time, relative to non-adopters. These are shown in Figure 7 and it is clear that adopters have larger employment, across all technologies.[5]

This exercise is repeated for productivity and blue-collar workers, but the results present a less consistent picture (see Figures B4 and B3 in Appendix D). There is little evidence of a differential trend in productivity for adopters relative to non-adopters, apart from for firms using Industrial Robotics. There is also no evidence of a significant deviation in blue-collar employment for technology adopters relative to non-adopters. These results suggest that technology adopters grow faster than non-adopters, but this occurs through hiring more skilled workers, rather than unskilled workers, where there is little evidence of change. This fits with the hypothesis of capital-skill complementarity (Krusell et al., 2000) with these advanced technologies, whereby adoption allows firms to grow by hiring more highly skilled workers.

It is important to note that these plots are only comparing adopters to non-adopters over time, but does not identify a causal effect from using these technologies. The interpretation is then clear - adopters seem to grow faster, especially from around 2014 onwards.

[5]There is reason to be concerned about endogeneity in these specifications: there are plenty of potential omitted variables which could be correlated with firm size, and technology adoption, such as a change in firm management. Therefore I present IV estimates in Appendix G, using investment expenditure on R&D over time as instruments. There is evidence of endogeneity, and instrument validity, but the estimates may suffer from weak instruments. The estimated coefficients for automation on employment are much larger, but should be treated with caution due to some evidence of weak instruments.

Figure 7. : Estimated β_t from TWFE model: Technology Adopters Grow Faster than Non-Adopters



Firms were asked *when* they started using various technologies, which allows me to identify the 'automation year'. This information helps me look at what happens to firms when they start automating. I estimate TWFE models as before, but the time-varying technology adoption variable permits both homogeneous and heterogeneous specifications.[6]

$$\ln Y_{it} = \alpha_i + \gamma_t + \delta X_{it} + \beta \mathbb{1}\text{Tech}_{it} + \epsilon_{it} \tag{2}$$

$$\ln Y_{it} = \alpha_i + \gamma_t + \delta X_{it} + \beta_t \mathbb{1}\text{Tech}_{it} \times \text{Year}_t + \epsilon_{it} \tag{3}$$

The results of these models are shown below, with the set of β in Table 6 and the series of β_t in Figure 8.

The results in Table 6 provide evidence that firms' employment rises upon adoption of each of these advanced technologies, in the range of 2 - 6 %. Although the point estimates for the effect of adoption on blue-collar employment are negative for Cloud Computing, AI & Big Data, and 3D Printing, the standard errors are large enough for the latter two that I cannot interpret these as statistically significant. Likewise, there is little evidence that using these advanced technologies raises turnover per worker, apart for Industrial Robotics and IoT. For robots, the estimate is large (at 6.5%) and statistically significant. The result for IoT is smaller (at 1.7%) and only holds at a 5% level of significance.

The plotted TWFE estimates in Figure 8 follow similar patterns to those in the previous

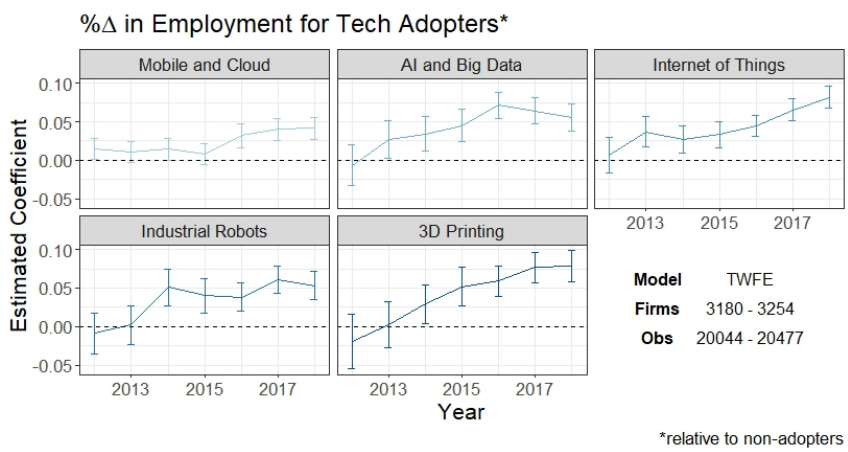
[6] As $\mathbb{1}\text{Tech}_{it}$ varies *within* firms over time, it is not perfectly correlated with the firm fixed-effect, so I can estimate a model which doesn't interact technology adoption and year.

Table 6—: Estimates of β from homogeneous effect TWFE model: the % change in variables when adopting technology, relative to non-adopters

		Cloud Computing	*AI & Big Data*	*IoT*	*Industrial Robotics*	*3D Printing*
Employment	Coeff	0.020***	0.052***	0.051***	0.042***	0.056***
	SE	(0.0043)	(0.0061)	(0.0048)	(0.0062)	(0.0066)
Blue-collar Emp.	Coeff	−0.036*	−0.030	0.0008	0.048	−0.025
	SE	(0.015)	(0.027)	(0.021)	(0.027)	(0.028)
Turnover per worker	Coeff	0.0057	−0.017	0.017*	0.065***	0.019
	SE	(0.0066)	(0.0096)	(0.0075)	(0.0097)	(0.010)

Notes: Robust standard errors clustered at firm level. Coefficients labelled by statistical significance at: *** 0.1%, ** 1%, * 5%.

Figure 8. : Estimates of β_t from heterogeneous effect TWFE model: Employment Grows When Firms Adopt Technologies



section, but the magnitude is slightly lower across the board. This suggests that accounting for the *timing* of adoption has only a small impact on the estimated impact of adoption, and thus selection effects are small. In other words, the employment impact of technology adoption can mostly be explained by the impact of moving from non-adopter to adopter.

Knowledge of *when* firms adopt technologies also permits an event study specification in which the employment series of firms is regressed on a set of dummy variables which indicate the time relative to adoption year. The specification takes the form:

$$\ln E_{it} = \mu_i + \delta_t + \gamma X_{it} + \sum_{j=\underline{j}, j\neq -1}^{\bar{j}} \beta_j \mathbb{1}(D_{it} = j) + \epsilon_{it}$$

where $D_{it} = t - A_i$ is the 'relative time', or the number of periods relative to when firm

i adopted technology in year A_i (Borusyak and Jaravel, 2017). The outcome of interest is log employment of firm i in year t. The regression includes both firm and year fixed effects. The time-invariant controls are sector and region, and the time-varying control is firm age.

The first set of results in panel (a) of Figure 9 simply show the coefficients β_j from the event study with the reference period the year before adoption. These results suggest that may be a linear pre-trend in log employment, especially for Industrial Robotics and 3D Printing. Therefore, I follow Borusyak and Jaravel (2017) and deal with this by identifying the deviation from a linear pre-trend. This is done by dropping two 'reference' periods, rather than just one. I drop the earliest period (6 years before adoption) and continue to exclude the period prior to adoption. The results are plotted in panel (b) of Figure 9, and although the linear pre-trends are still visible, they are shrunk towards zero and mostly not statistically significant.

The results from the event studies highlight a clear sustained rise in employment in the years following adoption of these advanced technologies. The boost to employment seems most pronounced for AI & Big Data, and 3D Printing, with the Internet of Things just behind. This is consistent with the previous results which considered the impact in adoption year using TWFE models.

More specifically, most of the point estimates prior to the reference year are insignificant across all technologies, apart from 5 years prior to adoption in some cases. There is also some mild evidence of a pre-trend in employment for adopters of IoT. Overall I am reasonably confident that the event study identifies how employment changes with technology adoption, with the caveat that I can probably better control for the linear pre-trends.

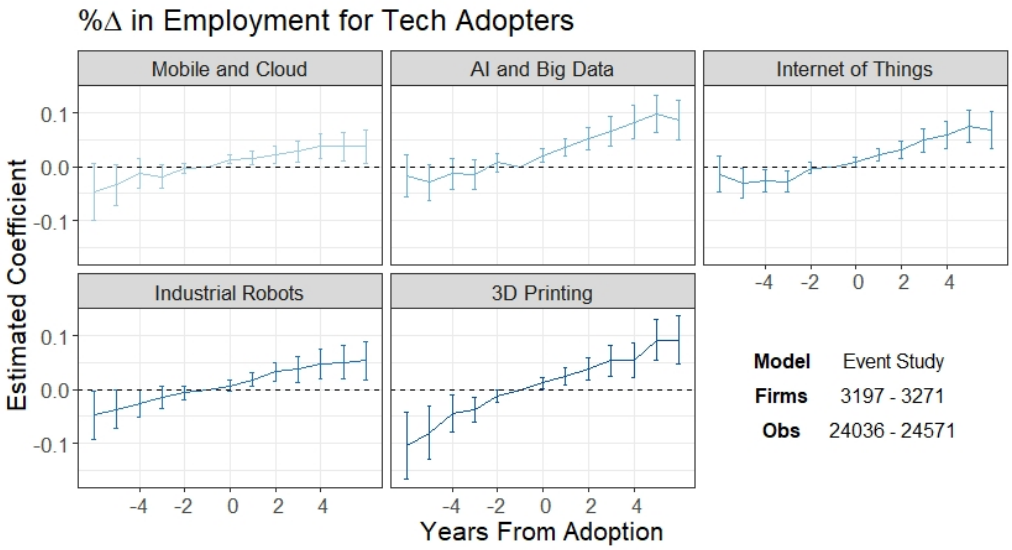
The results show a statistically significant relationship between technology adoption and firm size, in the range of 1 - 3 % in the first two years post-adoption, and rising to 4 - 11% in the final two years of the sample. These effects are statistically significant across all technologies, and the magnitudes are not trivial.

Conclusions

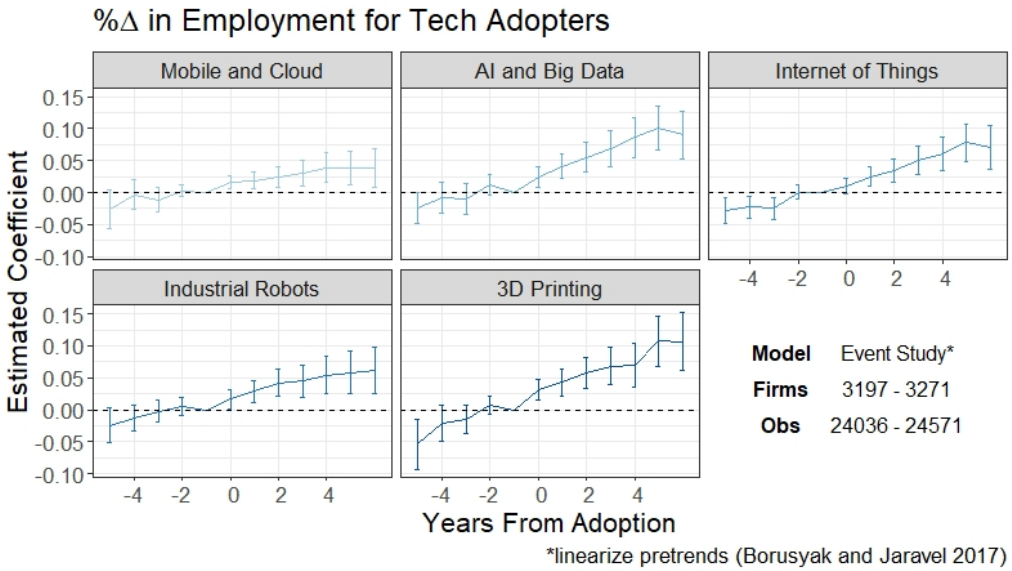
This empirical analysis covers plenty of ground, but there are some clear results: firms that automate tend to be larger, grow faster, hire more skilled workers, pay more, are more productive, and generate higher turnover. Identification of the relationship between technology adoption and firm outcomes is difficult, due to selection bias, so I do not claim that my estimates represent an accurate measure of the magnitude of causal effects. However, it does seem that firms using automation technologies seem to grow faster before adoption, but even more so *after* making the investment. Furthermore, automation seems to be somewhat reversible, as some firms adopted new technologies and subsequently dropped them. This set of facts is used to inform a model of heterogeneous firms which have the option to automate, introduced in the next section.

Figure 9. : Estimates from Event Study Regression

(a) Standard event study design, without controlling for pre-trends



(b) Event study design with approach to linearizing pre-trends as in Borusyak and Jaravel (2017)



IV. Model

I introduce a model of heterogeneous firms that can endogenously choose to automate. The aim of this model is twofold. Firstly, it allows me to analyse the aggregate impact of automation when firms differ and have the option to automate tasks, taking into account the trade-off between marginal and fixed costs from new technologies (De Ridder, 2019; Lashkari, Bauer and Boussard, 2019). Automation lowers marginal costs, making firms more productive, but there is selection in that more productive firms can afford the associated fixed costs. The impact on overall productivity is ex-ante unclear, and depends on both selection and reallocation. The aggregate effect on employment is also not obvious, as it depends on which firms expand, contract, enter, and exit. Secondly, I am interested in the general equilibrium effects. Automation influences the equilibrium price by affecting input costs. This will impact firm entry and exit. Furthermore, it has a labour market implications. The relative demand for skilled and unskilled workers will likely change, which benefits different firms in distinct ways.

I take a standard Hopenhayn (1992) heterogeneous firm dynamics model with adjustment costs on labour. I extend it to include a task-based production function (Zeira, 1998; Acemoglu and Restrepo, 2016), routine and non-routine labour which produce different sets of tasks, and automation technology which can replace routine workers.

Task-based Production Function with Automation

In the standard Hopenhayn (1992) model, firms are heterogeneous in productivity z and output is determined by one labour input n with decreasing returns to scale. The production function is $y = zn^{\alpha}$.

Taking inspiration from Acemoglu and Restrepo (2016), the new production function depends on productivity and production over a set of tasks x of increasing 'difficulty'. This allows for jobs that are made up of a variety of tasks, which might be performed by different inputs depending on their complexity. Thus firm output is:

$$\ln y = \ln z + \int_0^{\phi} \ln y(x) dx \quad \text{where } \phi < 1 \text{ for DRTS}$$

The production of each task $y(x)$ is determined by the set of tasks which can be performed by different inputs. I assume that routine labour n^r can be perfectly substituted by automation technology R, such as industrial robots, for some set of tasks. However, non-routine labour n^n cannot be substituted for, arcoss the more complex tasks. Of course these sets of tasks will change over time, as advanced technology improves. Production of

a task is determined:[7]

$$y(x) = \begin{cases} r(x) = R(x) + n^r(x) & \text{for } x \in [0, \gamma) \\ n^n(x) & \text{for } x \in [\gamma, \phi) \end{cases}$$

The prices of inputs are $\boldsymbol{w} = [w^n, w^r, q]$ for non-routine labour, routine labour and automation technology. This price q is the per-unit cost of using the technology, such as the electricity or computing power associated. I assume that $\frac{q}{w_r} < 1$ such that automating is beneficial. However firms must pay a fixed cost c_a in each period to use automation technology R. This can be thought of as the cost of paying a company to maintain the robots, or an annual payment to access cloud computing services. This does not scale with the amount of technology used, as does the technology input cost q. The production function is therefore:

$$\begin{aligned} &\ln y = \ln z + (\phi - \gamma) \ln n^n + \gamma \ln r \\ &\implies y = z(n^n)^\alpha r^\gamma \\ &\text{where } r = (n^r + R) \text{ and } \alpha = \phi - \gamma \end{aligned}$$

This yields a production function that has some capital-skill complementarity, which is an idea going back to at least Griliches (1969).

Heterogeneous Firm Model

Firms produce with decreasing returns to scale and choose inputs $\boldsymbol{x} = [n^n, n^r, R]$ to maximise profits:

$$\max_{\boldsymbol{x}} py(z, \boldsymbol{x}) - C(\boldsymbol{x}, \boldsymbol{w}) - g(n, n_{-1}) - c_f$$

where $n = n^n + n^r$ is the total labour input, n_{-1} is the previous period choice of labour inputs and $g(.,.)$ is an adjustment cost function. Firms differ in productivity z, which follows AR(1) process:

$$\ln z_{t+1} = \ln \bar{z} + \rho \ln z_t + \sigma_z$$

The value function of a firm is the largest of the 'automating' and 'non-automating' value functions, taking into account the fixed automation cost.

$$\begin{aligned} v_t^a(z_t, n_{t-1}) = \max_{R_t, n_t^n, n_t^r \geq 0} \{ & p_t z_t (n_t^n)^\alpha (R_t)^\gamma - w_t^n n_t^n - w_t^r n_t^r - q_t R_t - g(n_t, n_{t-1}) - c_f \\ & + \beta \max\{ \int v_{t+1}(z_{t+1}, n_t) dF(z_{t+1}|z_t), -g(0, n_t) \} \} \end{aligned}$$

[7] Choosing to split labour inputs into routine and non-routine is inspired by a rich literature on the substitutability of automation-related technologies with labour inputs. For example, Jaimovich and Siu (2012) look at the changes in employment shares by occupation group in the US, and find this distinction to be important.

$$v_t(z_t, n_{t-1}) = \max_{n_t^r, n_t^n \geq 0} \{p_t z_t (n_t^n)^\alpha (n_t^r)^\gamma - w_t^n n_t^n - w_t^n n_t^r - g(n_t, n_{t-1}) - c_f$$
$$+\beta \max\{\int v_{t+1}(z_{t+1}, n_t) dF(z_{t+1}|z_t), -g(0, n_t)\}\}$$

$$\tilde{v}(z_t, n_{t-1}) = \max\{v_t^a(z_t, n_{t-1}) - c_a, v_t(z_t, n_{t-1})\}$$

As in Hopenhayn (1992), there is a cut-off level of productivity z^* such that all firms with $z \geq z^*$ do not exit. Furthermore, there will be a cut-off level of productivity z^a such that all firms with $z \geq z^a$ will automate. The equilibrium price is pinned down by the free-entry condition $v^e(z) = \int_z v(z) dG(z) = c_e$. Finally the law of motion for the distribution of firms can be computed:

$$\mu_{t+1}(z') = \underbrace{\int_{z \geq z^*} \mu_t(z) dF(z'|z)}_{\text{Surviving incumbents}} + \underbrace{MG(z')}_{\text{Entrants}}$$

Solution Method

To solve this problem, I take first-order conditions (FOCs) of the static profit maximisation problem, for firms that automate, and firms that do not. These solutions can be found in Appendix I. The presence of the labour adjustment cost prevents a closed-form solution for optimal input demands, so this is solved numerically. A brief outline of the solution algorithm is presented here:

1) Numerically solve optimal input demands across grid of productivity and lagged employment.
2) Use the free-entry condition to find equilibrium price, both with and without the presence of automation technology.
    a) Guess price p_0
    b) Take FOCs of static firm problem to find maximised profit $\pi(p_0; z, n_{-1})$.
    c) Apply value function iteration on this starting point until convergence.
    d) Compute expected value of entry, and check free-entry condition holds.
    e) Adjust guess of price until free-entry condition holds.
3) Find cutoff productivity level for entry, and for automation.
4) Compute firm choices and distributions.

Calibration

The chosen labour adjustment cost function is quadratic, such that $g(n_t, n_{t-1}) = \frac{\tau}{2}(n_t - n_{t-1})^2$ where n_t is the sum of routine and non-routine labour hired in period t. The

purpose of this adjustment cost is to better match the data, as justified by long-standing literature (Kydland and Prescott, 1991; Hamermesh, 1989; Cogley and Nason, 1995). The labour adjustment cost is for *total* labour hired: firms can move workers from routine to non-routine tasks, and it is the hiring and firing of workers which incurs a cost.

	Parameter	**Value**	**Target**
pre-set			
β	Discount rate	0.95	Match annual IR over period
c_e	Entry cost	$0.82 \times c_f$	Barseghyan and DiCecio (2011)
$\frac{w_n}{w_r}$	Wage ratio	1.2	Vannutelli et al. (2019)
q	Automation price	0.99	By assumption
γ	Routine share	0.66 - α	Residual labour share
ρ	AR1 coefficient	0.8	
σ	AR1 stdev	0.2	
τ	Labour adjustment cost	0.5	Mendoza (1991)
calibrated			
α	Non-routine share	0.32	Match avg firm size
$\bar{z}$	AR1 intercept	1	Match avg labour productivity
c_f	Fixed cost	20	Match exit rate
c_a	Automation cost	8	Match % firms that automate
$\bar{D}$	Demand scalar	114.75	Normalise # firms to unity

The pre-set calibration parameters are fairly standard, or taken from existing literature. The automation price is set below the routine wage by assumption, for now. The initial calibrated parameters are chosen to target observables which are sensitive to that parameter. The non-routine share is chosen to match the average firm size, as this will determine the labour share in highly productive, automating firms. The AR(1) intercept is chosen to match the average labour productivity, as it determines the level of the productivity distribution from which firms draw. The fixed and automation costs are naturally chosen to match the exit rate and the share of automating firms.

The baseline calibration is achieved with a Genetic Algorithm, as in De Ridder (2019). This involves choosing parameters to find the global minimum of an objective function. In this case, I choose the calibrated parameters to minimise a weighted average of the absolute distance between the model and targeted moments, where the weights are the inverse of the targets themselves.[8]

Results

I report the following targeted and non-targeted moments in the model and data. The fit is reasonable, although the share of employment in automating firms is little low in

[8]Such weights force the algorithm to be 'unit-agnostic', else it would fit larger targets better than smaller ones.

the model, as a function of the perfect substitutability of automation technology and routine labour. I think that targeting the adjustment cost will help with this - with an appropriate labour adjustment cost, firms will be less likely to fully replace routine labour with technology, even if it is optimal without adjustment costs. This calibration exercise will be available in the next draft of the paper.

For the non-targeted moments, the routine share of employment of 57% is taken from Vannutelli et al. (2019), while the share of employment and output in automating firms is computed from the Italian survey data in the empirical section. For the targeted moments, the average firm size is from Schivardi and Torrini (2003). The exit rate and the average labour productivity are both taken from Eurostat, for Italy in 2016. The proportion of firms that automate is computed from the Italian survey data.

	Model	**Data**
Non-targeted		
Routine share of employment	0.64	0.57
Share of employment in automating firms	0.35	0.43
Share of output in automating firms	0.55	0.55
Targeted		
Avg firm size	8.20	3.90
Avg labour productivity	61.79	108.3
Exit rate	0.095	0.077
% firms that automate	0.37	0.30

Another exercise to check the model describes the data well is to calibrate the non-automated model as a starting point. To this end, I set the nonroutine to routine wage ratio at rate consistent with the data 'pre-automation' in the 1980s, at 1.1.[9] Then I back out the labour supplies implied by the model. Automation is introduced the model, and wages are backed out to clear the labour market. The nonroutine to routine wage ratio the model gives is 1.285, which is slightly higher than the data, but in the right direction. Overall it seems that this baseline model does a reasonable job of matching the data.

What is the Impact of Automation?

So what is the impact of introducing an automation technology in this model? Imagine that initially the per-unit price of automation q_t is greater than that of routine labour, or that automation technology is too undeveloped to produce any tasks that humans perform. The following table compares the model with and without the option to automate:

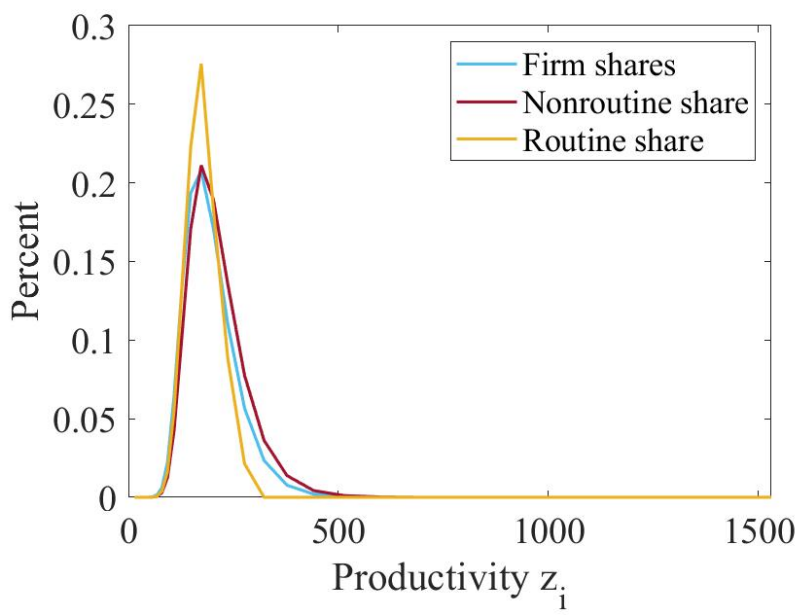
When cheaper automation technology is available to firms, they purchase this input *if they are productive*. This can be seen in the firm distribution Figure 10, where the routine labour share falls to zero for highly productive firms, as they fully automate this input.

[9] https://www.bancaditalia.it/pubblicazioni/qef/2019-0478/QEF_478_19.pdf

Table 7—: Percentage point change relative to 'No Automation' model

Aggregates:	Price	-3.2
	# firms	-9.0
	Output	+3.7
	Employment	-14.9
	Output-weighted productivity	+2.9
	Total profit	+28.9
	Share routine employment	-6.0
Firm Level:	Employment per firm	-21.9
	Output per firm	-6.7
	% firms that automate	+37.0

Figure 10. : Employment Shares in Model with Automation



These firms are thus able to expand due to the low-cost input, and earn significantly greater profits. The general equilibrium effect is a lower price, because automating firms can produce output more cheaply. This reduces returns to low-productivity firms, so they exit. Overall, the reallocation towards more-productive firms leads to a rise in output-weighted productivity. Workers suffer from a large fall in employment, and this is skewed towards routine labour which takes up a smaller share of total employment. At a firm level, the average firm is smaller and produces less, due to the greater skew of output

towards highly productive firms. However, firms that automate are larger, as they are able to expand with the new technology.

We can thus reconcile the aggregate findings (Acemoglu and Restrepo, 2018; Dauth et al., 2021) that automation technology reduces employment, with my firm-level evidence that automaters grow on adoption. This model highlights the important role for heterogeneity, reallocation, and general equilibrium effects. There are two important extensions which are being worked into the next version of this paper. Firstly, I will introduce an endogeneous labour supply, to allow the labour market channel to affect firm behaviour. Secondly, I want to quantitatively assess the balance of automation on directly boosting productivity, versus reallocating towards more-productive firms.

V. Conclusion

This paper has sought to understand the adoption of automation technologies among firms, and the general equilibrium effects thereof. On the empirical side, I take a novel dataset on firm use of advanced technologies, and I show that automating firms tend to be larger, but also grow faster, both before and after they adopt. The employment growth doesn't show up in blue-collar workers, suggesting that complementarities between automation technologies and skilled workers are strong and important. Furthermore, adopters of new technologies are more productive, but I do not find that productivity is boosted by adoption.

Taking stock of these findings, I extend Hopenhayn (1992)'s seminal heterogeneous firm model to include two types of labour - routine and nonroutine - alongside automation capital which can be substituted for the former. In the model, automation is achieved only by highly productive firms, due to the fixed cost of using this technology. These firms benefit most from automation, allowing them to expand. The general equilibrium effect lowers prices, forcing low-productivity firms out of the market. Unsurprisingly, workers lose out significantly, especially those in routine occupations. The critical next step is to allow wages to respond to changing demand and supply for labour. The implication is that routine wages will fall more than nonroutine wages, leading to adjustments to firm demand for these factors, which will have nontrivial effects on incentives to automate, entry, exit and aggregate distributional consequences.

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Appendix A

The following section contains detailed analysis on adopters of automation technologies contained in the Bank of Italy's "Survey of Industrial and Service Firms".

There are five distinct automation technologies in the survey: Mobile & Cloud Computing, Artificial Intelligence & Big Data, Internet of Things, Industrial Robotics, 3D Printing. Firms are asked about adoption of these technologies in 2015 and 2017. In this section I look at each technology separately and investigate the firm characteristics associated with adoption. Associated figures and tables are included at the end of this section.

I run logit regressions for technology adoption on a wider range of variables than in the main body of the paper. For each technology, in 2015 and 2017, I run the following specification on a large set of potential covariates, including time-invariant characteristics and time-varying firm choices and outcomes:

$$\begin{aligned} \mathbb{1}\text{Tech}_{i,t} &= \beta_0 + \beta_1 \text{Sector}_i + \beta_2 \text{Region}_i + \beta_3 \text{Age}_{i,t} + \beta_4 \text{Exporting Bin}_{i,t} \\ &+ \beta_5 \ln(\text{Avg. Wage})_{i,t} + \beta_6 \ln(\text{Turnover})_{i,t} + \beta_7 \text{Fixed I}_{i,t} + \beta_8 \text{Hires}_{i,t} \\ &+ \beta_9 \text{Terminations}_{i,t} + \beta_{10} \text{Employment}_{i,t} + \beta_{11} \text{Blue-Collar Employment}_{i,t} + \epsilon_{i,t} \end{aligned} \tag{A1}$$

Where $\mathbb{1}\text{Tech}_{i,t}$ is a binary variable indicating if firm i adopted the technology in year t, and the independent variables are self-explanatory. I then employ stepwise variable selection using the Akaike Information Criterion (AIC) to select a parsimonious model that can best explain the factors related to technology adoption. The results will be discussed in detail in each subsection. Briefly, the estimated coefficients give the marginal contribution to the likelihood of adopting each technology. As expected, I find heterogeneity across industries and regions. Generally, it seems that larger firms are more likely to use these advanced technologies. In addition, I often find that more hirings increases the probability of adoption, while more firings does the opposite. This suggests that it is not only the size of the firm that is associated with technological adoption, but whether it is growing, or not. The final consistent finding across all technologies is that adoption is positively related to turnover. The full set of results are in Tables A3 and A4 at the end of this section, and show the estimated coefficients and included variables in each model.[10]

Mobile and Cloud Computing

In 2015, the survey combined Mobile Broadband and Cloud Computing together, whereas the latter was isolated in 2017. Mobile Broadband is quite a widely-defined technology, and the survey notes this by giving specific examples encompassing wireless technology, high-speed internet and mobile devices (see Appendix B). Cloud Computing is not narrowly defined in the survey, and can encompass a range of technologies such as software

[10] No standard errors nor p-values are given, because p-values in stepwise regression models are biased. However, this information is not particularly meaningful here; these models simply include the variables that best explain technological adoption. Included variables are considered valuable in this regard.

(e.g. Google Docs), platforms (e.g. Microsoft Azure), and data storage (e.g. Amazon S3) (Bayrak, Conley and Wilkie, 2011).

I find that in Italy a large proportion of firms use Mobile and Cloud Computing: 73% say they use this technology in 2015. For Cloud Computing specifically, it is unsurprisingly much lower, at 21% in 2017. It is more common in larger firms, but adoption doesn't seem to vary with firm age.[11] The strong positive relationship between use and size is even more stark when just looking at Cloud Computing in 2017, and this relationship holds across all age groups.

Adoption of Mobile & Cloud Computing seems to be less common for heavy exporters in 2015, but this pattern doesn't seem to hold for just Cloud Computing in 2017. This can be seen in Figure A2, which plots adoption of this technology by export share.

The sectoral distribution of Cloud Computing is quite equal, as can be seen in Figure A3. For example, while approximately 21% of firms use Cloud Computing in 2017, the sectors in which it is most commonly adopted - Real Estate and Transport & Communications - are at around 30%. Cloud adoption sits between 15 - 20% for most other sectors, with low uptake only in Hotels & Restaurants.

Firms using Mobile Broadband and the Cloud tend to have a significantly lower proportion of workers in blue collar jobs, and higher fixed investment per worker (see Table A1 at the end of this section). Adopters of this technology are also more active in the labour market, with much higher terminations and hirings per permanent employee compared to non-adopters.

Cloud Computing is the only technology in this survey (other than E-Commerce, which is also only asked about in 2017) more commonly adopted in Services firms compared to other industries. Around 30% of firms in Real Estate, Transport and Communications are using this technology. It is perhaps unexpected that adoption is still above 10% across all Manufacturing sectors.

From the logit regressions (Equation A1), I find that industry is important in explaining adoption across all technologies in 2015. For Mobile and Cloud, firm adoption is more common in Hotels & Restaurants, and Transport & Comms., but less so in Minerals Processing and Textiles Manufacturing. More concretely, firms are almost 10 times more likely to adopt this technology if they work in Hotels & Restaurants compared to Food, Beverages & Tobacco.[12] The estimated coefficients for the other variables suggest that the likelihood of adoption increases around 0.3% with each additional worker, falls around 0.3% with each extra blue-collar employee, rises around 14% with a unit rise in log(Turnover), and falls 0.1% with each year the firm ages. Finally, it seems that heavy exporters are less likely to adopt Mobile & Cloud technology. If a firm exports more than two thirds of its sales, the probability of technological adoption falls by over 40% compared to non-exporters. This finding is consistent with the simple averages plotted in Figure A2.

For firms using only Cloud Computing technology in 2017, there are some similarities: a high export share is associated with a lower likelihood of adoption ; while adopting

[11] This actually contradicts one of the few firm-level analyses of Cloud adoption, which suggests that firm size does not matter (Benlian, Hess and Buxmann, 2009)

[12] From Table A3, $e^{2.279} \approx 9.77$.

Cloud is increasing in employment, turnover and hirings. However, the industry patterns are quite different, which reflects a significant difference between Cloud Computing and Mobile. In addition, the probability of using Cloud Computing is increasing in firm age, at about 0.3% per year.

Artificial Intelligence and Big Data

Firms that use AI & Big Data are younger, larger, have higher turnover per worker, and higher fixed investment. On average, firms adopting this technology have 63 workers and are 33 years old, compared to 49 workers and a mean age over 35 for non-adopters. Importantly, these differences are statistically significant.[13]

Approximately 11% of firms in the sample are classified as using AI & Big Data in 2015. When firms are binned by employment, the difference in AI use is stark, as can be seen in Figure A4. For firms with under 200 workers, around 10% of them use AI. It is closer to 25% of firms that employ over 200 workers. Comparing the age distribution in the same way presents a less clear pattern, although there is some evidence that younger firms tend to adopt AI and Big Data.

The fact that a larger proportion of big firms use AI & Big Data also holds when the data is split by age and size. Figure A6 shows the proportion of AI firms in age $\times$ size bins - it doesn't seem that AI differs by firm age, but it does vary by employment. This can be seen more clearly in Figures A5, which show AI and Big Data use by employment (age), split into lines by age (employment).

AI & Big Data firms also have a much lower proportion of workers in blue collar jobs, with just 43% compared to 55% in firms that don't use this technology. In fact, of all technologies in this dataset, AI & Big Data firms have the lowest share of manual labourers by some margin (the next lowest is 50.1% for firms using Internet of Things). This could speak to the complementarity of high-skilled workers with this advanced technology.

Adoption of AI & Big Data is far more common in Services, such as Transport, Retail and Communications, ranging from 12 - 21%. On the flip side, adoption is below 5% in Food, Drink, and Textiles manufacturing firms. Adoption across all industries can be found in Figure A7.

Firms using AI & Big Data also have a peculiar relationship with exporting. It is only this technology for which firms export significantly *less*, whether we compare by average share of sales in exports (see Table A1) or looking at the distribution, as in Figure A8.

Results from the logit regression in equation A1 can shed further light, as they show (jointly) the important variables associated with technology adoption. Unsurprisingly, it seems that AI & Big Data is more commonly adopted in Services compared to Manufacturing. Firms operating in Transport & Comms. are over 7 times more likely to adopt AI & Big Data than those in Food, Beverages & Tobacco.[14]

The results from 2015 support the findings that adopters export less: the probability of adoption is 20 - 30% *lower* for firms exporting more than a third of their sales, relative

[13] At the 1% level, computed with Welch's t-test and the Welch-Sattherwaite equation for degrees of freedom.
[14] From Table A3, $e^{2.028} \approx 7.60$

to non-exporters. Once I partial out the effect of industry, region and export status, the likelihood of adoption rises by 0.03% with each additional worker, and falls 0.04% with each extra blue-collar employee. Furthermore, AI & Big Data adoption rises around 30% and 40% with a unit rise in log(Wage per worker) and log(Turnover), respectively. To the extent that wages and turnover are correlated with labour productivity, this suggests that this is somewhat associated with adopting AI & Big Data. Finally, I find that probability of adoption falls around 0.6% with each year firms age, conditional on the other factors in the model. This supports the finding that younger firms are more likely to adopt this technology.

The logit results are qualitatively the same in 2017, with slightly different magnitudes, which don't meaningfully change the interpretation. The only significant difference is that adoption is now more likely for firms that export a large share of sales. One potential explanation is that AI & Big Data technology allows firms to scale up quickly and export, which tends to be associated with firm size (Love and Roper, 2015).

Internet of Things

Firms using IoT technologies are indistinguishable in their age profile from those that do not adopt, both in the average age (see Table A1) and age distribution (see Figure A9). However, once again, it seems that larger firms are more likely to adopt this advanced technology: the average employment of IoT firms is almost 62 workers, compared to 48 for non-adopters. This pattern is consistent in 2015 and 2017, with the larger firms more likely to adopt IoT technology across all age groups, as in Figure A11.

Furthermore, there are some noticeable differences in the characteristics of firms using IoT between 2015 and 2017, which may not be surprising for a novel technology. In 2015, there doesn't seem to be any pattern in IoT adoption by either exporting behaviour or macro industry. By 2017, it seems clear that a very small proportion of non-exporters use IoT technology compared to exporters. This can be clearly seen in Figure A10. Furthermore, there is growth in IoT use among firms in manufacturing and mining, with little change in adoption for businesses in services.

Finally, there is differing IoT adoption by region, with a higher uptake for firms in the North-East. This likely reflects sectoral differences, with around 20% of those in Metal Manufacturing and Transport & Communication adopting the technology, compared to under 10% in Textiles Manufacturing and Hospitality. IoT firms have a significantly lower proportion of workers in blue-collar occupations, and exports contribute a smaller percentage of total sales, compared to non-adopters.

When partialling out industry effects, I find that the likelihood of adoption of IoT is consistently lower for exporters. Firms exporting over 1/3 of their sales are 30% - 40% less likely to use IoT compared to non-exporters.[15] Furthermore, it seems that adoption of IoT is 0.1% more likely with each year firms age, and 0.006% less likely with each additional worker. As with other technologies, firms are significantly more likely to use this technology if wages per worker or turnover are higher. The results are broadly similar

[15] From Table A3, $e^{-0.548} \approx 0.578$ and $e^{-0.334} \approx 0.716$.

in 2017, although the marginal contribution of age and wages have flipped signs. This suggests that, by 2017, younger firms and those with lower wages per worker were more likely to adopt IoT.

Industrial Robotics

I find that adopters are significantly larger and older than firms that don't use robots. These patterns can be clearly seen in Figure A12. In fact, the difference in both the average age, and the average size, between adopters and non-adopters is the largest out of the technologies considered in this paper. Firms using industrial robots have a mean age of over 41, compared to under 35 for non-adopters. In addition, firms adopting industrial robotics employ over 68 workers on average, compared to just 48 employees for non-adopters.

Furthermore, this relationship holds for the joint distribution of age $\times$ size, as can be seen in Figure A13, with the mass clustering in one corner.

Firms using industrial robots pay more, on average, and have a higher proportion of blue-collar workers, compared to non-adopters (see Table A1). In fact, industrial robot adopters have 64% of their workforce in blue-collar occupations, which is by far the highest of any of the technologies in this survey (the next highest is firms using 3D printing, with 56%).

The sectoral distribution of robot adoption is, unsurprisingly, extremely unequal: robots are used by around 25% of Metal Manufacturing firms, but by barely any businesses in Services. This distribution is plotted in Figure A15.

It is rare that non-exporting firms use robots. For example, in 2017 adoption was under 2.5% for domestic-only firms, compared to around 15% adoption for firms with over a third of sales coming from exports. This stark distribution can be seen in Figure A16, which plots the share of firms using robots by their export share. However, the average (employment-weighted) proportion of sales coming from exports is actually *lower* for robot adopters, with just 5% compared to 10% for non-adopters. This suggests that robot adopters that export are relatively small, by employment, compared to non-adopters. This intuitively makes sense - if robots displace jobs, they allow firms to upscale and export with less of a rise in employment. I find that this hypothesis is supported by the descriptive evidence. Consider Figure A14, which plots the proportion of robot adopters in each employment bin for four levels of exporting behaviour. It is clear that robot adoption rises with firm size for exporting-focused firms (exporting more than a third of sales), but this relationship disappears for firms that export up to one third of sales.[16] This means that the proportion of *these* exporting firms that use robots does not depend on employment, so robot-exporters are relatively smaller than their technologically-lagging counterparts.

The technology adoption model of equation A1 finds that use of Industrial Robotics in 2015 is explained by firm industry, region, export status, turnover and wage per worker.

[16] Around 31.5% of firms are in the bucket that do export, but it contributes to less than one third of sales.

Obviously, firms using industrial robots are very unlikely to operate in Services. Interestingly, once industry effects are controlled for, the probability of using robots seems to rise with export share. Firms are over 16% more likely to adopt industrial robots if they export over two thirds of their sales, compared to non-exporters.[17] Adoption likelihood is *decreasing* in log(Wage per worker), with a fall of almost 40% for each unit rise in this variable. A priori we might expect firms to be more likely to purchase robots if their wages are high, in terms of replacing relatively expensive workers. Therefore, it is perfectly reasonable to think that firms which have adopted robots will pay lower wages, on average, as they have possibly already displaced high-wage workers. Finally, I find that the likelihood of robot adoption falls around 0.007% with each additional worker. Again, this is consistent with the displacement hypothesis: robot-adopters are smaller (conditional on industry, region, turnover etc.) because they have been able to replace workers with machines.

The results from the technology adoption model in 2017 are similar across industries, regions and export status. I continue to find that robot adoption is positively associated with turnover. However, the model suggests that the probability of using robots *rises* with firm size (at 0.2% per worker). It seems that the conditional probability of adopting robots flipped from depending negatively on firm size in 2015, to positively in 2017.

3D Printing

The survey used in this research allows me to highlight facts about firm adoption of 3D printing that, to my knowledge, have not yet been investigated. In 2015 I find that adopting firms are older and larger. However, by 2017, firms using 3D printing are still larger, but the age distribution is much flatter. It seems that fast-growing young firms adopted this technology in recent years. This is suggestive evidence that 3D printing allows young firms to grow very quickly. Plots of these patterns can be found in Figure A18.

Similarly to robotics, it is rare than non-exporting firms use 3D printing. For example, less than 2% of non-exporting firms adopt this technology, compared to over 14% of 'heavy' exporters (those exporting over two-thirds of sales). This pattern is seen clearly in Figures A19. However, I do not find the same result that, on average, adopters export a smaller proportion of their sales than non-adopters. For 3D printing firms, the share of sales in exports is indistinguishable from other firms (see Table A1).

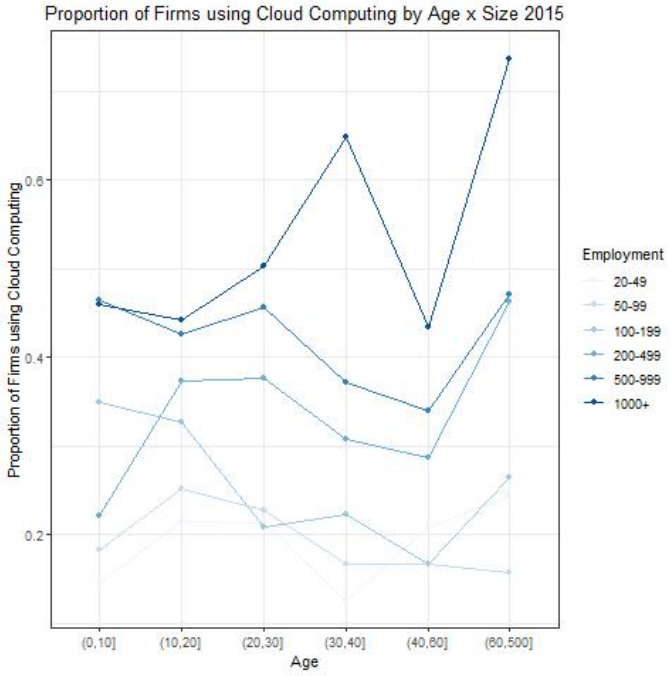
As with industrial robots, 3D printing is much more common in Manufacturing than other sectors, and it is especially concentrated in Metal and Rubber, Chemical & Plastics (see Figures A20). Analysis of the factors related to adoption of 3D Printing technology using equation A1 underscore the higher likelihood of adopting in Manufacturing over Services: firms are over 20 times more likely to use 3D Printing if they are in Metal Manufacturing compared to Food, Beverages & Tobacco production.[18]

Furthermore, once sectors are controlled for, adoption of 3D Printing is more likely for exporters, and less likely in the North-West of Italy. There is evidence that the probability

[17] From Table A3, $e^{0.151} \approx 1.163$.
[18] From Table A3, $e^{3.146} \approx 23.243$.

of adoption rises with firm size, with an increase of 0.01% for the marginal employee. Like all the other technologies in this paper, adoption seems to rise with firm turnover, being 10% more likely with a unit increase in log(Turnover). These findings hold in 2017: the likelihood of adoption rises with employment at 0.02% per worker, and rises at 11% with a unit rise in log(Turnover). However, in 2017 I find that 3D Printing is less likely for exporters compared to non-exporters. If the aforementioned hypothesis that fast-growing young firms adopted this technology in recent years, and such firms are less likely to export (Love and Roper, 2015), then this finding supports the theory.

Figure A1. : Cloud Computing More Common in Larger Firms



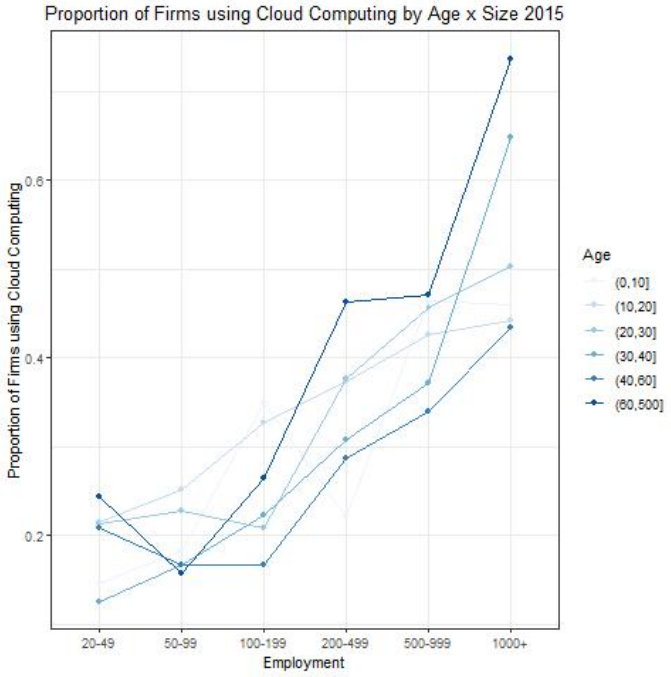


Figure A2. : Mobile & Cloud Adoption Falls with Exporting Share (2015), but Use of Cloud Computing Alone Does Not (2017)

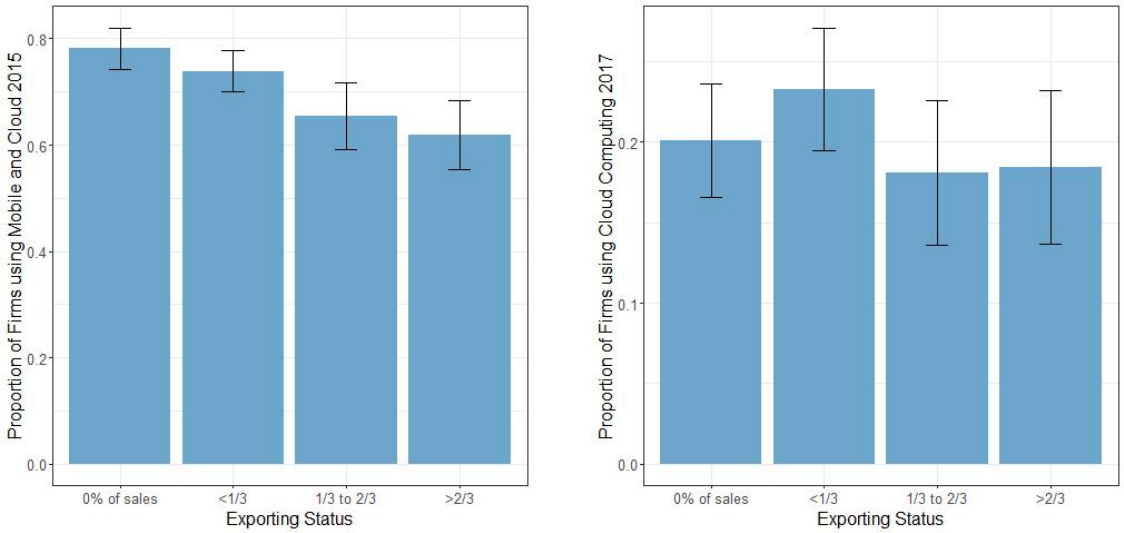


Figure A3. : Cloud Computing Adoption Varies Across Industries (2017)

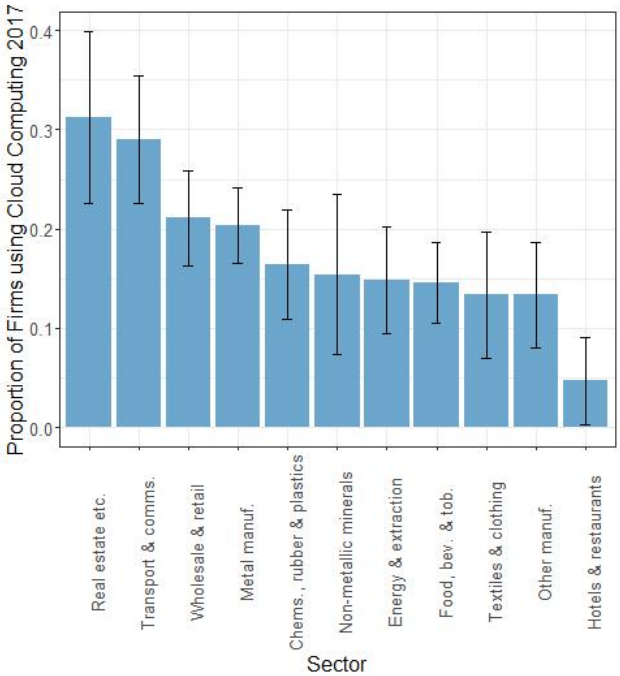


Figure A4. : AI & Big Data More Common in Larger Firms, Somewhat in Younger Firms (2015)

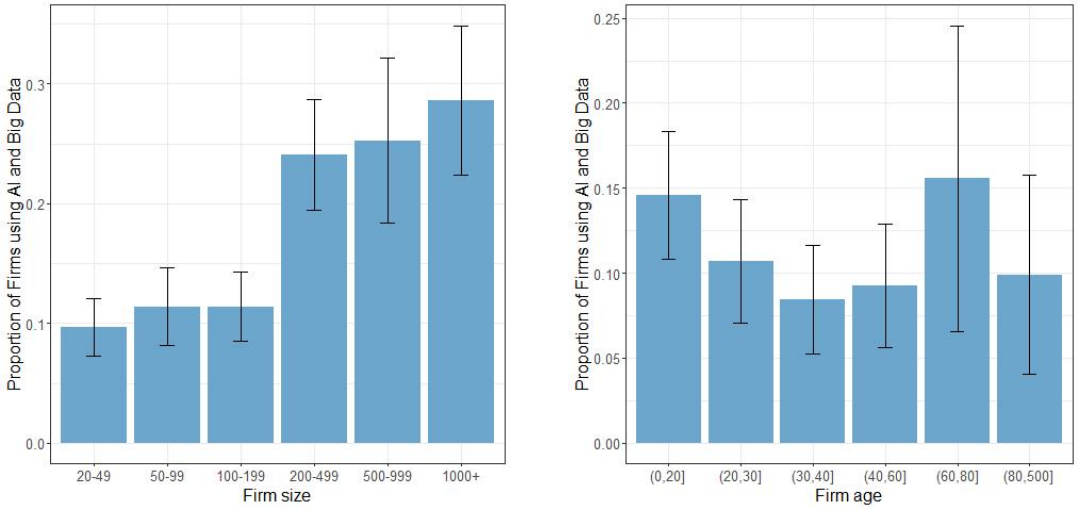


Figure A5. : Age-size Distribution 2015: Proportion of firms using AI & Big Data

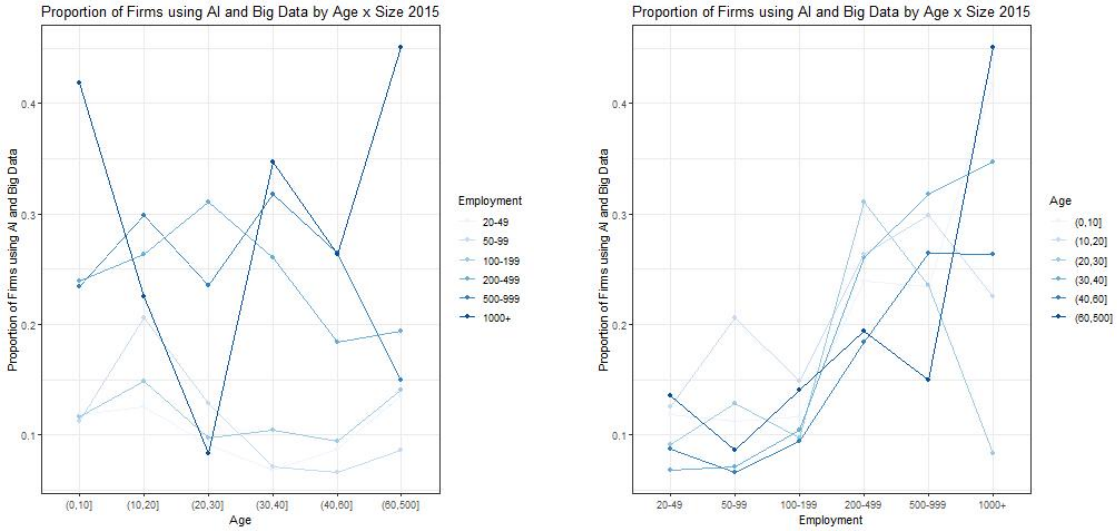


Figure A6. : Age-size Distribution 2015: Each bar is ratio of firms using AI & Big Data in cell

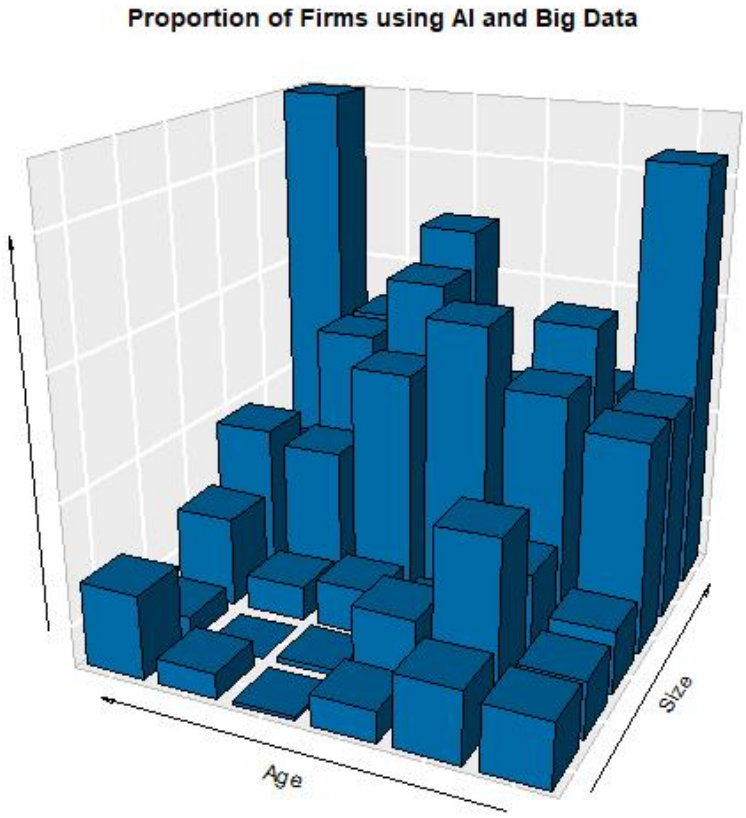


Figure A7. : Sectoral Variation in Adoption of AI & Big Data (2015)

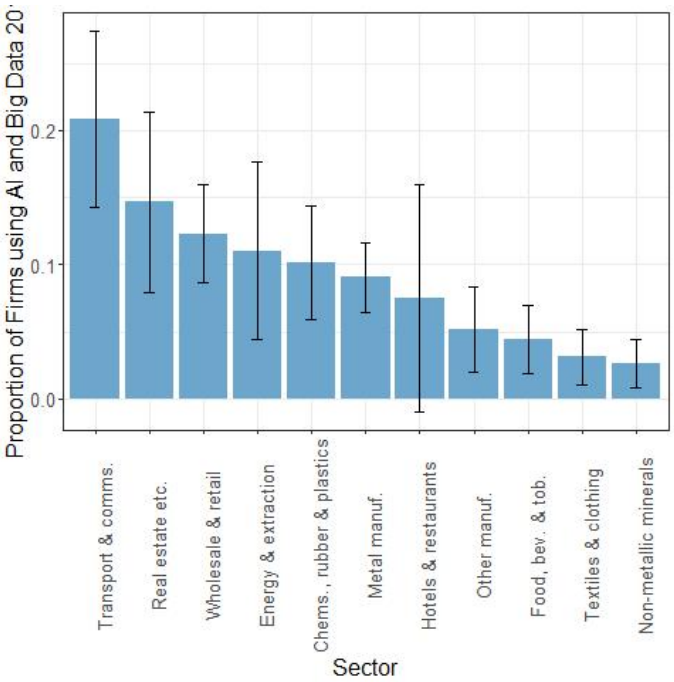


Figure A8. : Smaller share of exporting firms use AI & Big Data

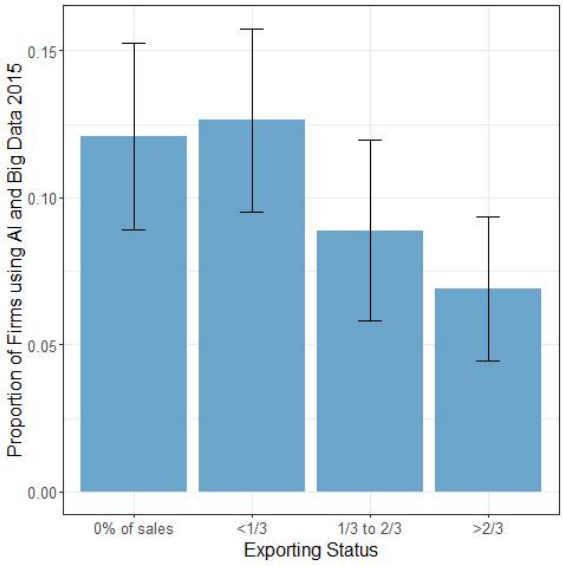


Figure A9. : Internet of Things More Common in Larger Firms, Doesn't Vary with Firm Age (2015)

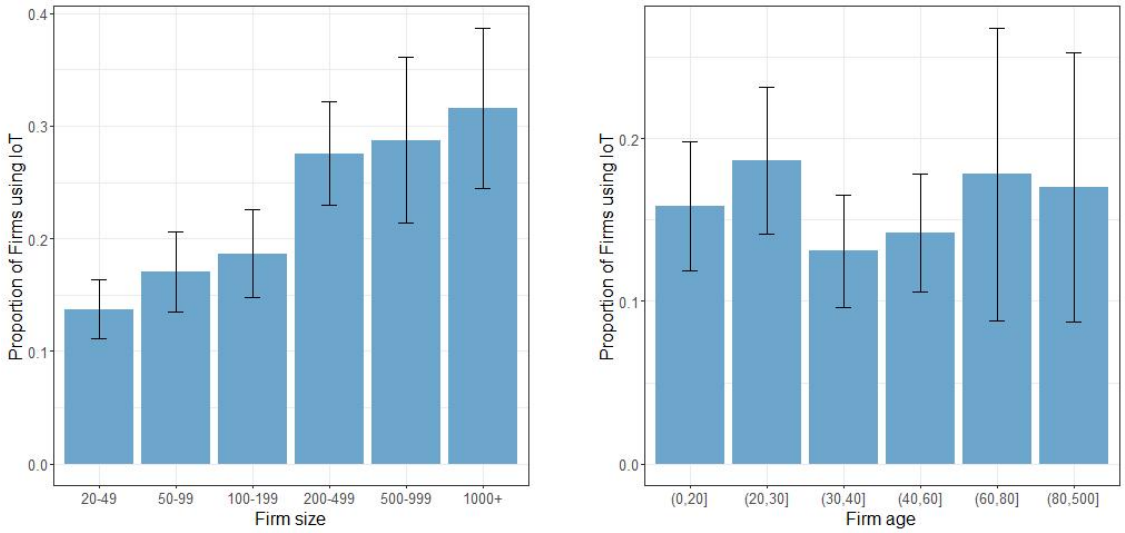


Figure A10. : Adoption of Internet of Things More Common For Exporters by 2017

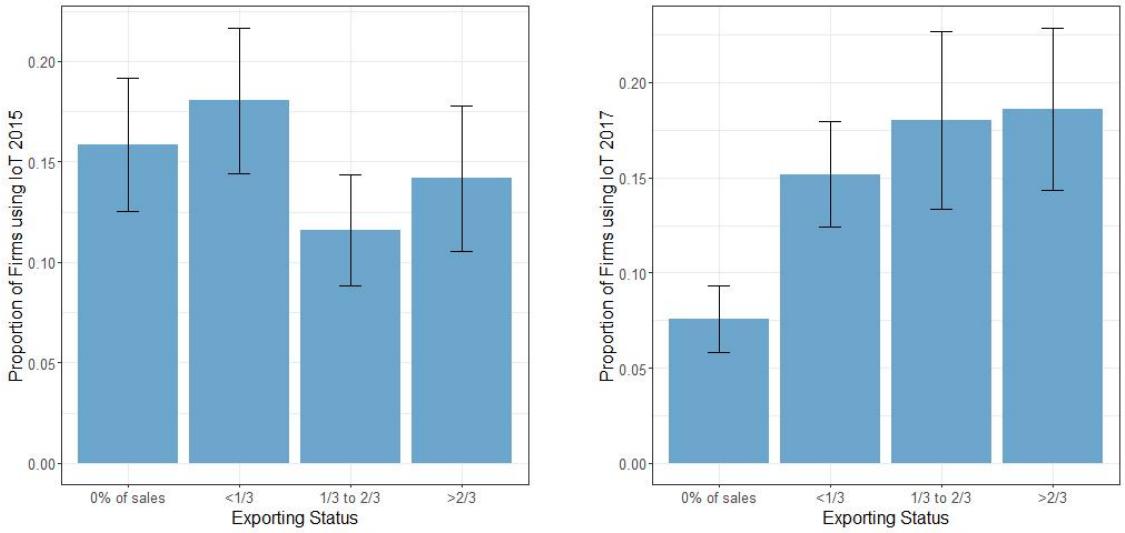


Figure A11. : Internet of Things adoption rises with size across all age groups

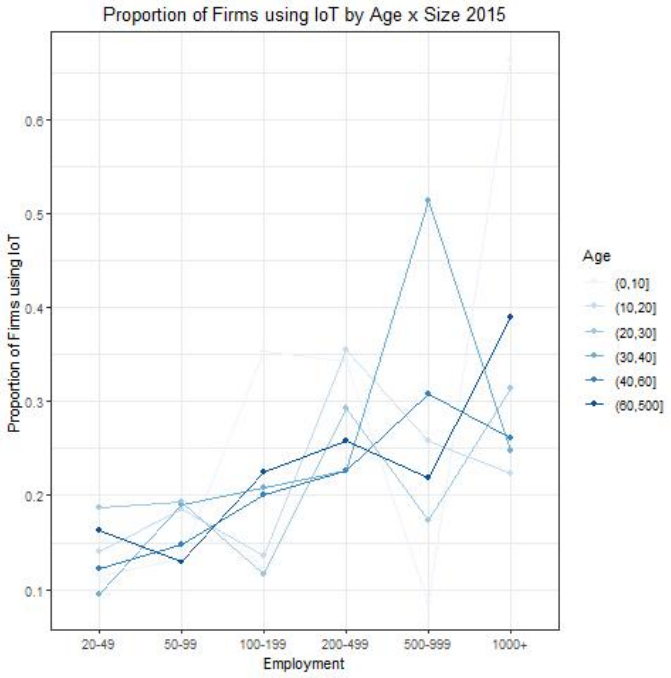


Figure A12. : Industrial Robotics More Common in Larger, Older Firms (2015)

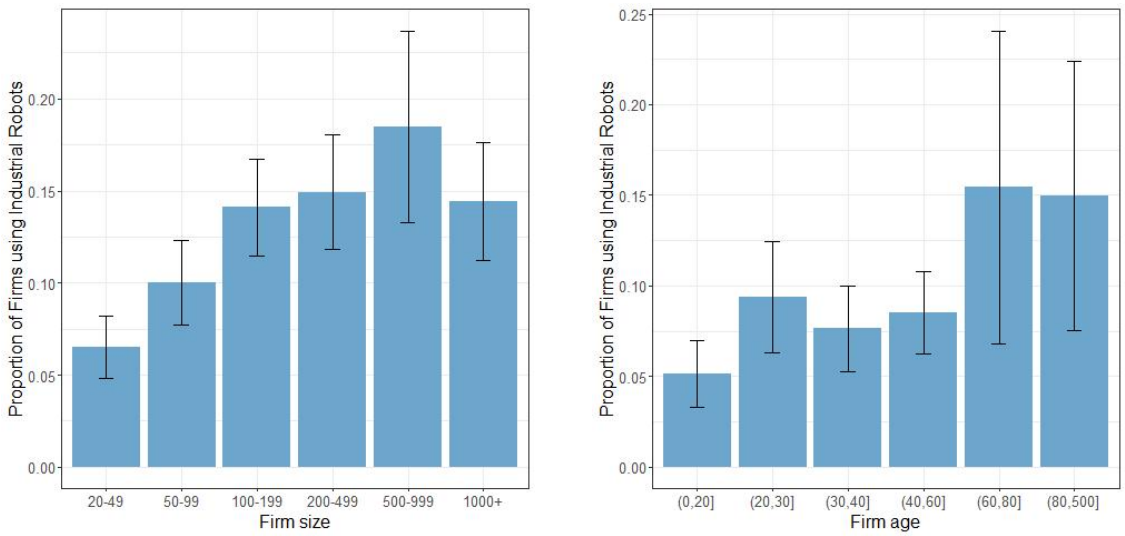
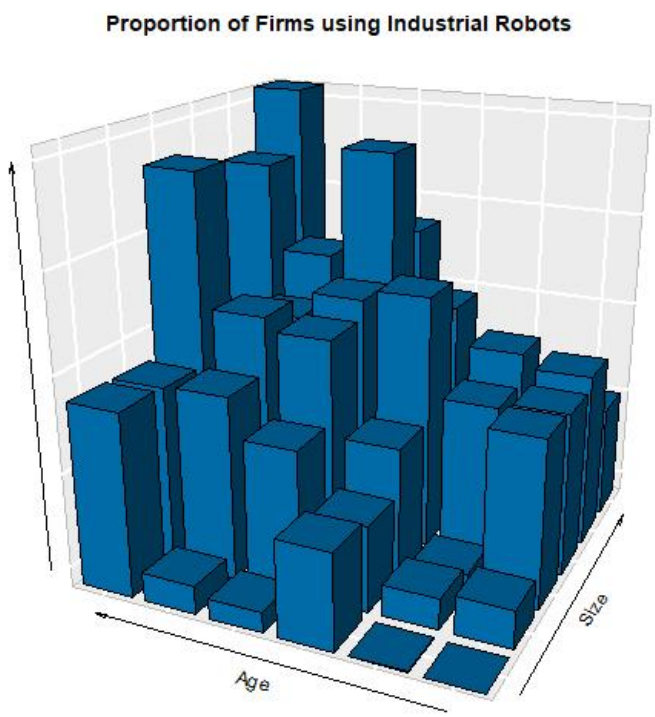


Figure A13. : Age-size Distribution: Each bar is ratio of firms using Industrial Robots in cell



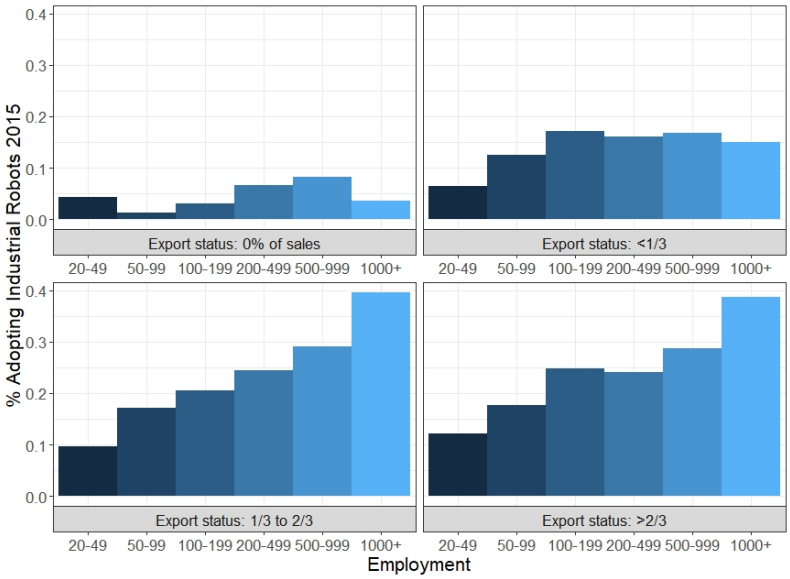


Figure A14. : Industrial Robots Adoption by Employment and Exporting Status, 2015

Figure A15. : Adoption of Industrial Robotics is Concentrated in Manufacturing (2017)

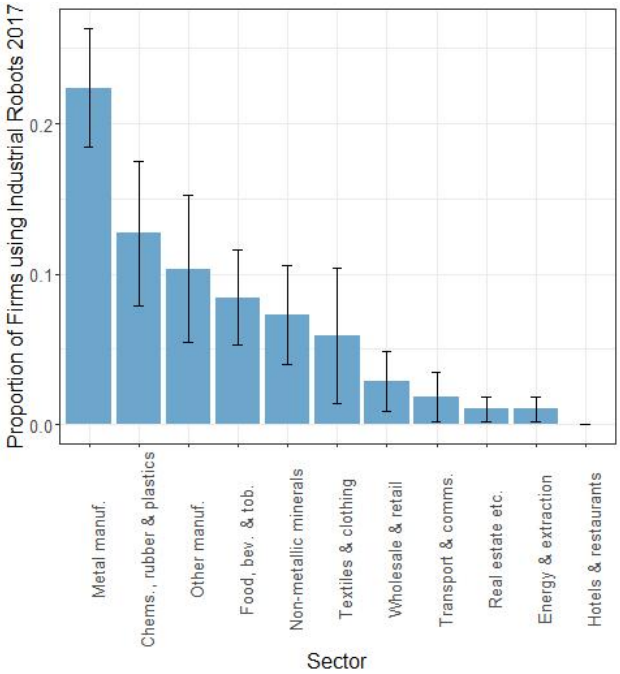


Figure A16. : Industrial Robotics Adoption Much More Common For Heavy Exporters

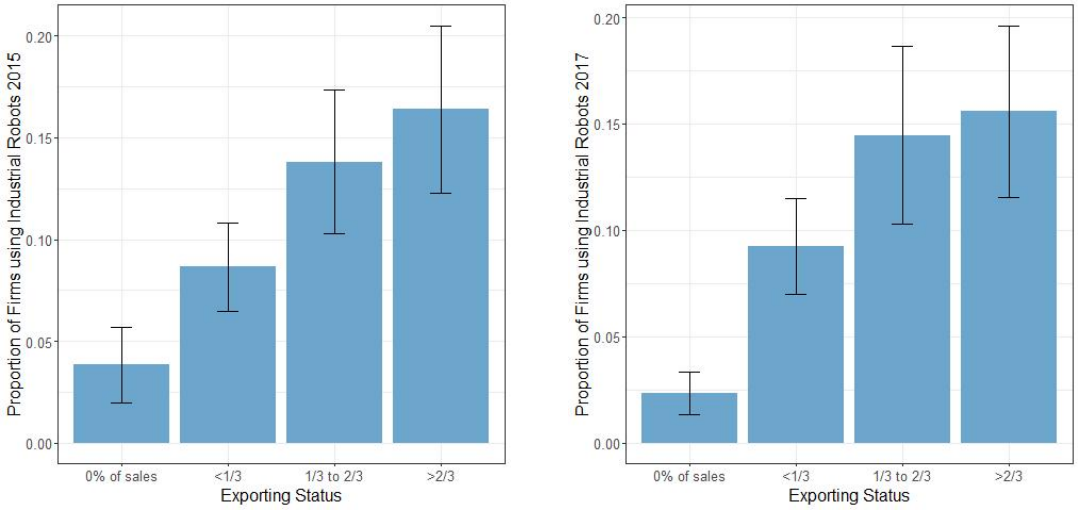


Figure A17. : 3D Printing Adoption Rises with Firm Size and Age, 2015

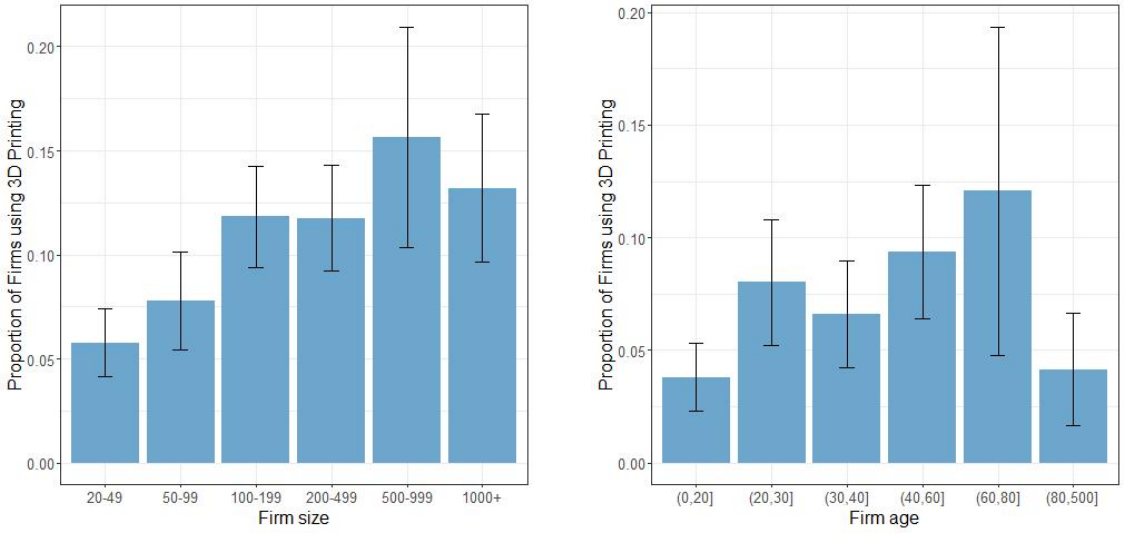


Figure A18. : 3D Printing Adoption Rises with Firm Size, but not Age, 2017

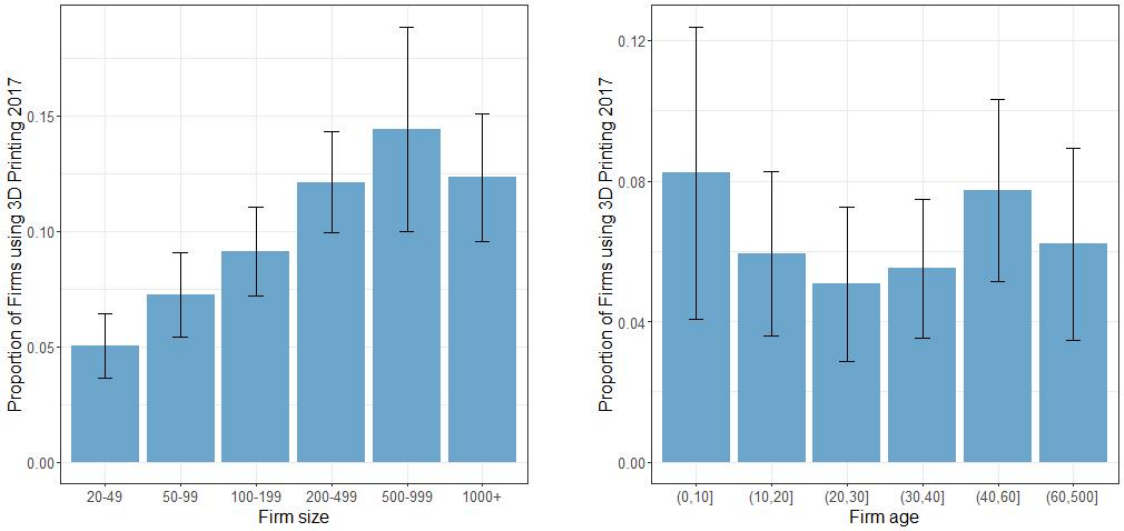


Figure A19. : 3D Printing Adoption Much More Common For Heavy Exporters

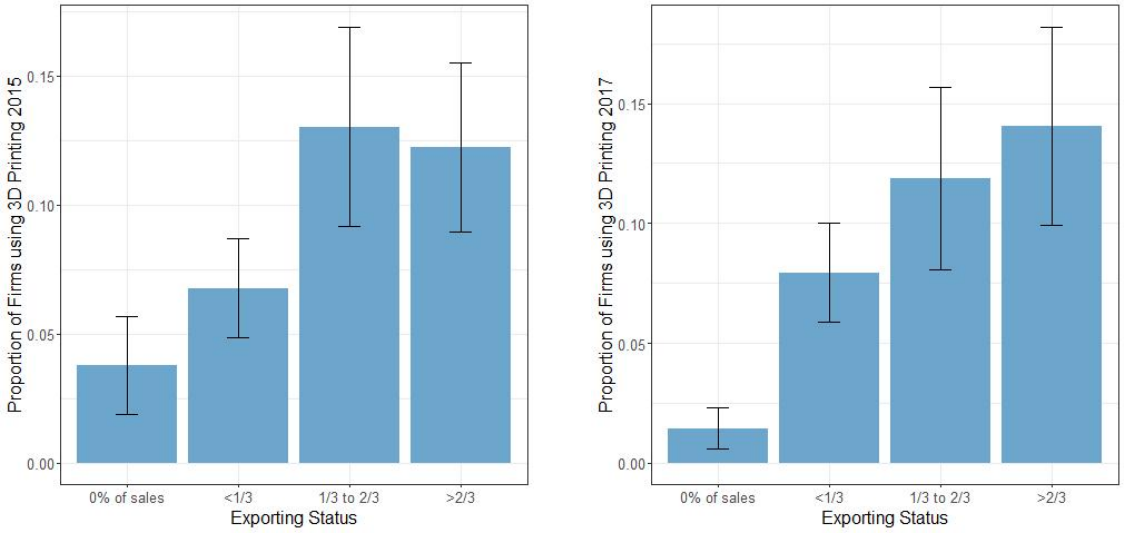
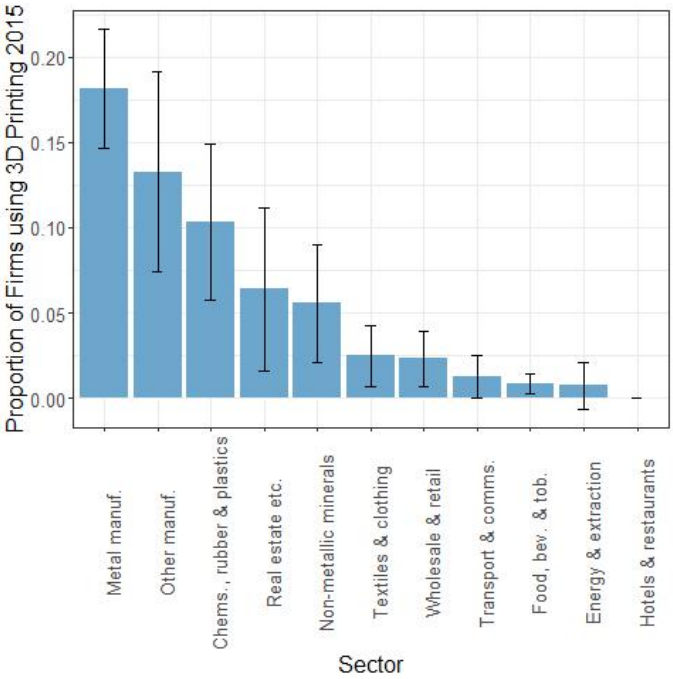


Figure A20. : Adoption of 3D Printing is Concentrated in Manufacturing



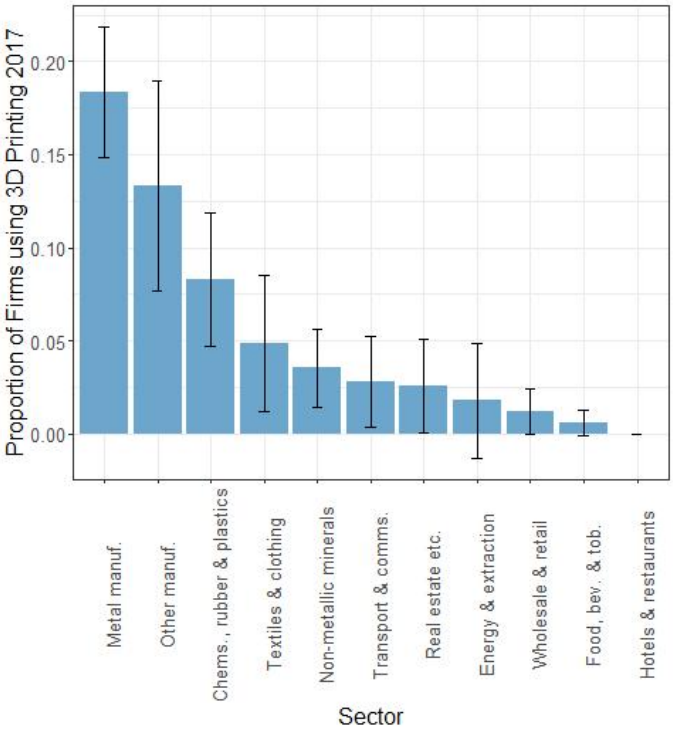


Table A1—: Advanced Technology Adopters vs Non-Adopters: Summary Statistics 2015

	Mobile & Cloud		*AI & Big Data*	
	Adopters	**Non-Adopters**	**Adopters**	**Non-Adopters**
Age	34.9	**35.7**	33.0	**35.4**
Log(employment)	**3.94**	3.83	**4.14**	3.89
Log(wage per worker) (€)	**10.28**	10.28	**10.35**	10.27
Hours per worker (weekly)	**31.46**	31.40	**31.65**	31.49
Turnover per worker (thou., €)	**459**	345	**499**	423
Fixed investment per worker (thou., €)	**10.9**	8.12	**14.5**	9.61
% blue-collar	0.51	**0.61**	0.43	**0.55**
% export sales	0.09	0.11	0.06	**0.10**
Terminations per worker	**0.18**	0.14	0.17	0.17
Hirings per worker	**0.19**	0.13	0.18	0.17
Number of firms	2524	934	440	2953
	Internet of Things		*Industrial Robotics*	
	Adopters	**Non-Adopters**	**Adopters**	**Non-Adopters**
Age	35.1	35.1	**41.1**	34.7
Log(employment)	**4.13**	3.87	**4.22**	3.88
Log(wage per worker) (€)	**10.32**	10.27	**10.33**	10.28
Hours per worker (weekly)	**31.81**	31.46	31.05	**31.56**
Turnover per worker (thou., €)	397	**439**	331	**442**
Fixed investment per worker (thou., €)	**12.4**	9.78	8.22	**10.4**
% blue-collar	0.51	**0.55**	**0.64**	0.53
% export sales	0.04	**0.11**	0.05	**0.10**
Terminations per worker	0.16	0.17	0.11	**0.17**
Hirings per worker	0.18	0.18	0.11	**0.18**
Number of firms	642	2742	485	2912
	3D Printing			
	Adopters	**Non-Adopters**		
Age	**38.5**	34.9		
Log(employment)	**4.16**	3.89		
Log(wage per worker) (€)	**10.33**	10.28		
Hours per worker (weekly)	**31.65**	31.52		
Turnover per worker (thou., €)	273	**444**		
Fixed investment per worker (thou., €)	5.71	**10.5**		
% blue-collar	0.56	0.54		
% export sales	0.11	0.10		
Terminations per worker	0.08	**0.17**		
Hirings per worker	0.11	**0.18**		
Number of firms	360	3037		

Note: Summary statistics from 2015 for firms that do and don't use advanced technologies. All values (other than number of observations) are weighted means. Bold values are the larger of the two, if there is a significant difference between adopters and non-adopters at the 1% level, computed with Welch's t-test and the Welch-Sattherwaite equation for degrees of freedom.

Table A2—: Advanced Technology Adopters vs Non-Adopters: Summary Statistics 2015

	Mobile & Cloud		*AI & Big Data*	
	Adopters	**Non-Adopters**	**Adopters**	**Non-Adopters**
Fixed Investment per worker (thou., €):				
Buildings	**4.22**	1.80	**6.00**	3.18
Plant & machinery	6.98	**7.43**	**9.97**	6.79
Transport equipment	**0.32**	0.23	0.20	**0.30**
Number of firms	2524	934	440	2953
	Internet of Things		*Industrial Robotics*	
	Adopters	**Non-Adopters**	**Adopters**	**Non-Adopters**
Fixed Investment per worker (thou., €):				
Buildings	2.71	**3.69**	2.10	**3.67**
Plant & machinery	**8.01**	6.86	**7.72**	6.93
Transport equipment	0.26	**0.30**	0.18	**0.32**
Number of firms	642	2742	485	2912
	3D Printing			
	Adopters	**Non-Adopters**		
Fixed Investment per worker (thou., €):				
Buildings	1.82	**3.66**		
Plant & machinery	6.23	**7.26**		
Transport equipment	0.18	**0.31**		
Number of firms	360	3037		

Note: Summary statistics from 2015 for firms that do and don't use advanced technologies. All values (other than number of observations) are weighted means. Bold values are the larger of the two, if there is a significant difference between adopters and non-adopters at the 1% level, computed with Welch's t-test and the Welch-Sattherwaite equation for degrees of freedom.

Table A3—: Logit Regression: Technology Adoption 2015

	Mobile & Cloud	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
Age	−0.001	−0.006	0.001	0.002	0.003
log(Wage per worker)	-	0.252	0.199	−0.453	-
Employment	0.003	0.0003	-	−0.00007	0.0001
Blue-collar Emp.	−0.003	−0.0004	−0.00006	0.0002	0.0004
log(Turnover)	0.129	0.353	0.278	0.311	0.098
Investment	-	0.0000008	0.000003	-	−0.000007
Hirings	0.005	-	-	-	0.006
Terminations	−0.003	-	-	-	−0.008
Industry:					
SS2	−0.033	−0.333	0.135	−0.896	0.712
SS3	0.086	1.076	0.935	−0.567	2.300
SS4	−0.339	−0.535	0.030	0.267	1.956
SS5	0.176	0.975	1.509	0.326	3.146
SS6	0.168	−0.301	0.945	−0.257	2.927
SS7	0.303	0.922	0.846	−0.657	−12.480
SS8	0.532	0.900	0.804	−3.218	0.626
SS9	2.279	0.517	0.338	−16.280	−2.723
SS10	0.935	2.028	1.201	−1.462	0.710
SS11	0.648	1.543	1.179	−2.107	0.814
Region:					
North-East	0.593	0.093	0.316	-	0.341
Centre	0.106	−0.156	0.001	-	0.316
South & Islands	0.428	0.523	0.659	-	0.385
Export Share:					
$0 < x \leq 1/3$	0.076	0.214	−0.050	−0.179	0.319
$1/3 < x \leq 2/3$	−0.122	−0.217	−0.548	0.244	0.984
$x > 2/3$	−0.570	−0.394	−0.334	0.151	0.714

Note: Estimated coefficients from AIC stepwise selection logit model, regressing technology adoption in 2015 on industry, region, export status, age, wage, employment, turnover, investment, hirings and terminations. Regressions weighted by CAF adjustment Dargatz and Hill (1996), as discussed in Section **??**. Estimates for industry are with respect to the omitted industry SS1, for region with respect to the omitted region North-West, for export share with respect to non-exporters, and for firm size to small (20 - 49) firms. No statistical significance given due to bias of p-values in stepwise regression.

Table A4—: Logit Regression: Technology Adoption 2017

	Cloud Computing	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
Age	0.003	−0.005	−0.003	-	-
log(Wage per worker)	-	0.304	−0.900	−0.184	0.357
Employment	0.0003	0.0005	0.0001	0.0002	0.0002
Blue-collar Emp.	-	-	-	-	-
log(Turnover)	0.204	0.262	0.394	0.0323	0.104
Investment	-	0.000005	0.000006	-	-
Hirings	0.002	0.001	-	0.003	0.005
Terminations	−0.002	−0.002	−0.0007	−0.003	−0.006
Industry:					
SS2	0.250	0.353	−0.558	−0.481	2.262
SS3	0.187	0.260	0.076	0.151	2.802
SS4	0.399	0.372	−0.348	0.667	2.880
SS5	0.392	0.518	0.340	0.710	3.490
SS6	0.494	0.284	−0.331	0.066	3.548
SS7	0.702	0.531	0.191	−1.707	−0.599
SS8	0.381	−0.111	−0.773	−1.694	0.035
SS9	−0.658	−0.033	−1.432	−16.530	−12.680
SS10	1.107	0.847	0.213	−1.208	1.184
SS11	1.287	1.125	−1.192	−1.778	2.631
Region:					
North-East	0.594	-	0.131	−0.011	−0.109
Centre	−0.437	-	−0.517	0.246	−0.383
South & Islands	−0.388	-	−0.177	0.321	−0.153
Export Share:					
$0 < x \leq 1/3$	0.234	0.877	0.336	0.948	−0.109
$1/3 < x \leq 2/3$	−0.059	0.303	0.457	0.733	−0.383
$x > 2/3$	−0.199	0.476	0.239	0.854	−0.153

Note: Estimated coefficients from AIC stepwise selection logit model, regressing technology adoption in 2017 on industry, region, export status, age, wage, employment, turnover, investment, hirings and terminations. Regressions weighted by CAF adjustment Dargatz and Hill (1996), as discussed in Section **??**. Estimates for industry are with respect to the omitted industry SS1, for region with respect to the omitted region North-West, for export share with respect to non-exporters, and for firm size to small (20 - 49) firms. No statistical significance given due to bias of p-values in stepwise regression.

Appendix B

This section includes supporting figures and tables from analysis on the Bank of Italy's "Survey of Industrial and Service Firms".

Table B1—: Estimate of relationship between technology adoption and productivity

		Cloud Computing	*AI & Big Data*	*IoT*	*Industrial Robotics*	*3D Printing*
2015	Coeff	0.087*	0.103	0.095*	−0.019	−0.070
	SE	(0.046)	(0.100)	(0.059)	(0.074)	(0.080)
2017	Coeff	0.120**	0.169**	0.121**	0.067	0.055
	SE	(0.056)	(0.069)	(0.054)	(0.054)	(0.068)

Notes: Robust standard errors computed. Coefficients labelled by statistical significance at: *** 1%, ** 5%, * 10%.

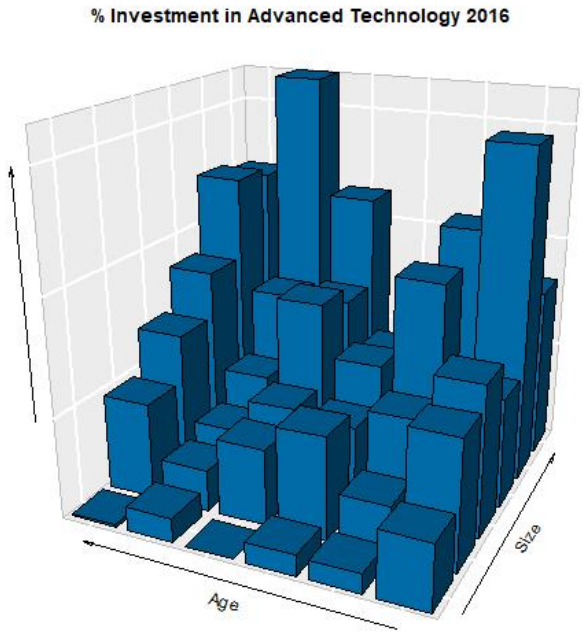
This table contains estimated coefficients from standard firm growth regressions, on log employment and age, with year and firm fixed-effects. The regressions are run over all firms in the panel that appear in more than two years between 2010 - 2018.

Table B2—: Estimated Coefficients from Firm Growth Regressions

Dependent variable: *Employment Growth*								
log(Emp.)	0.001	0.002	0.001	0.002	−0.384***	−0.384***	−0.383***	−0.384***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.09)	(0.09)	(0.09)	(0.09)
Age		−0.0003*		−0.0003*		0.001***		0.0006
		(0.0001)		(0.0001)		(0.0003)		(0.0003)
Year FE			✓	✓			✓	✓
Firm FE					✓	✓	✓	✓
N	20450	20446	20450	20446	20450	20446	20450	20446

Note: Estimates are significant at levels of 0.1%: ***, 1%: **, 5% *.

Figure B1. : Firm Age × Size Distributions of Advanced Technology Spending



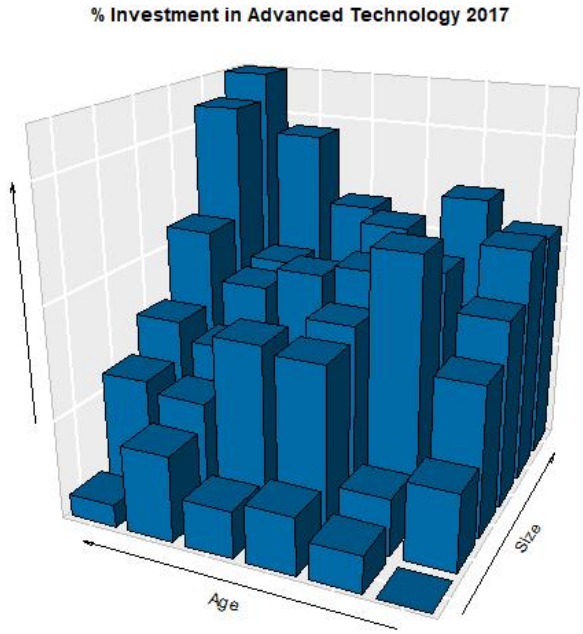
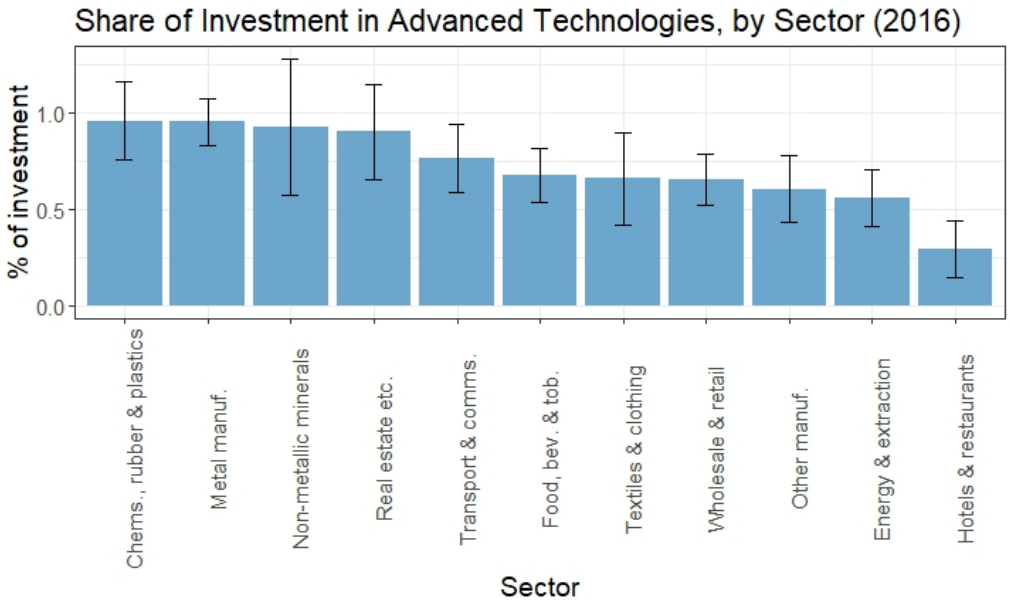


Figure B2. : Sectoral Distribution of Advanced Technology Spending



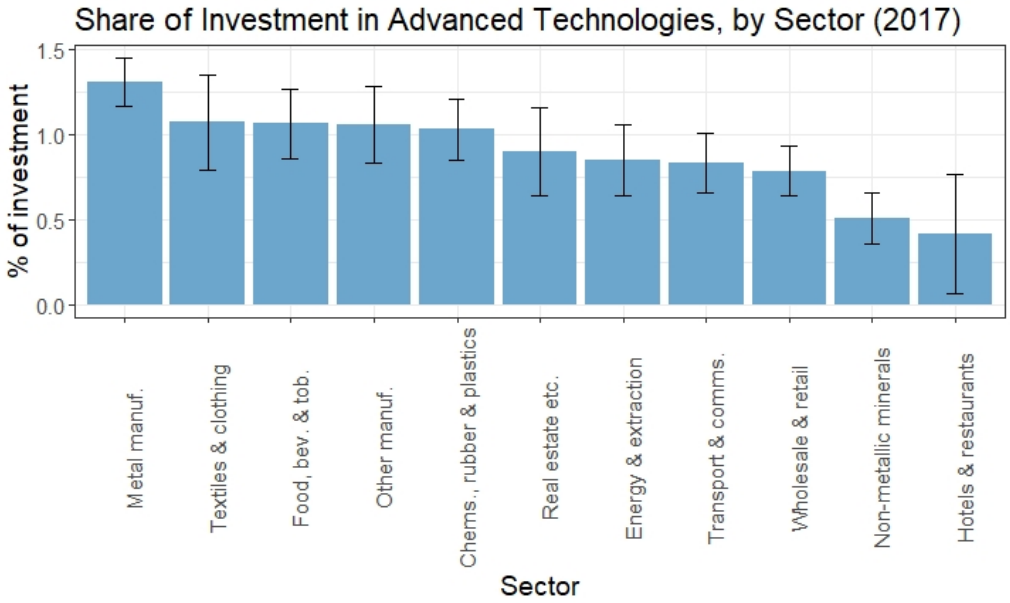


Figure B3. : Estimated β_t from TWFE model: Little Evidence of Deviation in Blue-Collar Employment for Adopters

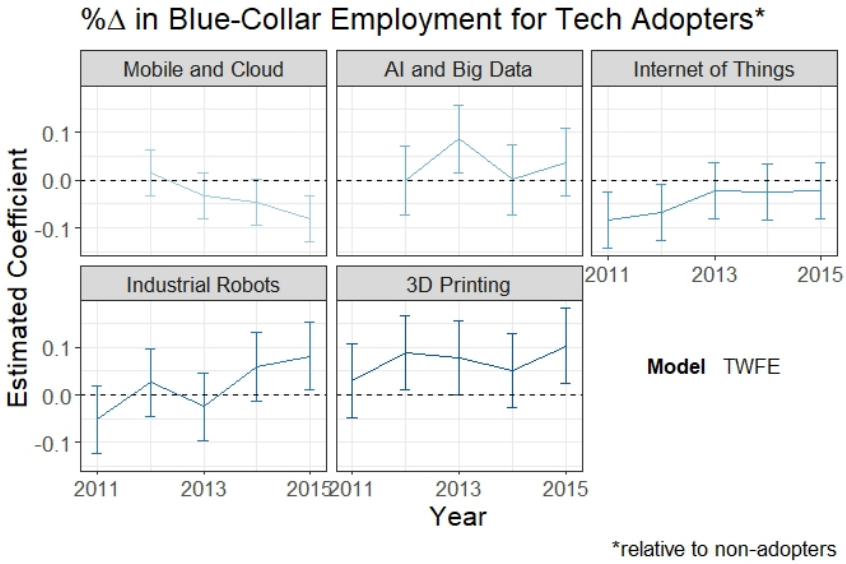


Figure B4. : Estimated β_t from TWFE model: Technology Adopters' Productivity Doesn't Rise Faster than Non-Adopters

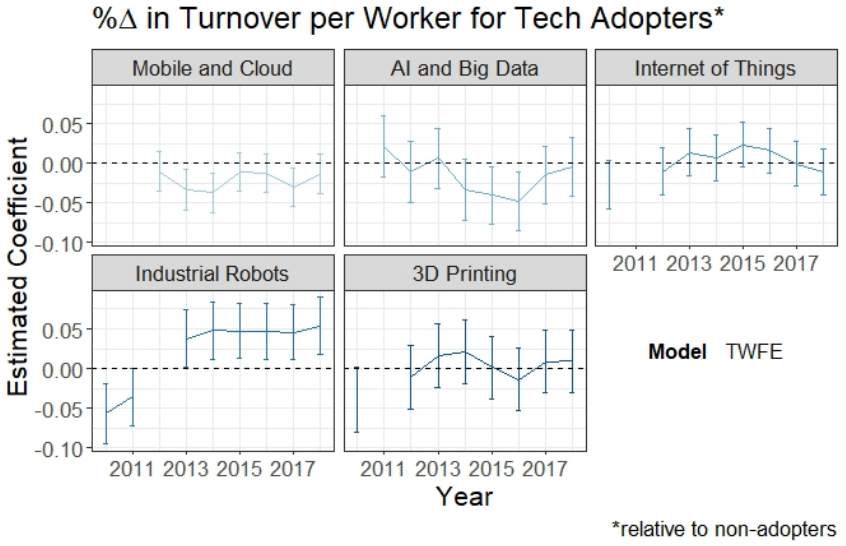
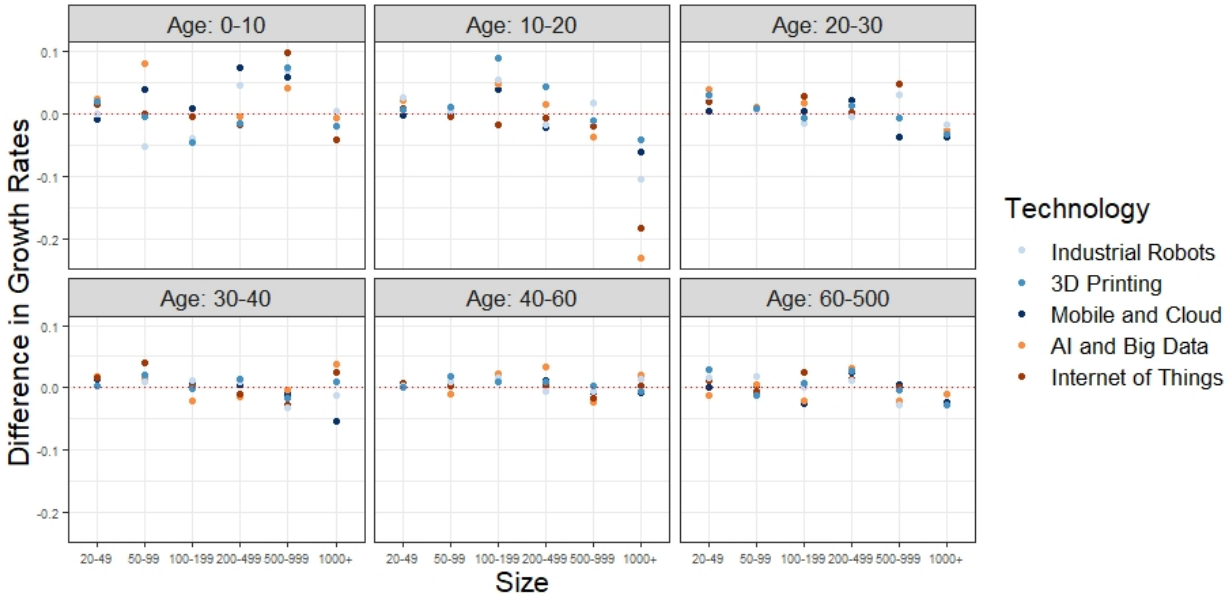


Figure B5. : Average Growth Rates (2010 - 2018) for Automation Adopters vs Non-Adopters, By Size and Age



Appendix C

This section includes further details on the Bank of Italy's "Survey of Industrial and Service Firms".

Survey Design

[19] The survey collects annual data on firm employment, investment, turnover, debt, and other structural information. It also has further detail on specific issues, which may not be asked annually, such as strategies, governance, technological and organisational factors. Up until 1998, only manufacturing firms with more than 50 employees were covered, but this was expanded in 1999 to include extractive and energy firms. Firms with more than 20 workers were included from 2002.

The survey population is divided into strata and firms are chosen randomly from each for the sample (one-stage stratified sample). Each strata are defined by economic activity, firm size, and region of head office. Employment is measured as the average number of workers during the year. Firms included in the previous year of the survey are always contacted, but may not be kept in the sample if they change activity class or fall below the threshold number of employees (they are replaced by firms of the same industry and size).

The data undergoes extensive quality checks: answers must fit within the range for the question; the panel data must be consistent; outliers must be checked. Statistical methods such as 'selective editing' are used to summarise the impact of outliers, to reduce the requirement to re-contact firms.

Advanced Technologies Questions

The automation technology questions are contained here in full. In 2015, firms were asked if they use or intend to adopt the following advanced technologies:

1) Mobile broadband and the cloud (e.g. wireless technology, apps, smartphones, tablets, high-speed broadband, and cloud management software)

2) Artificial intelligence and big data (e.g. collection and use of large data sets that, with the application of specific algorithms for machine learning, can provide support to decision making; possible applications are: in remote access diagnostics, defining algorithms for financial investments, patent-related or legal searches)

3) The internet of things (e.g. the use of technologies that, by means of advanced sensors, allow apparatus to be used in the production and commercial processes promoting their integration)

4) Industrial robotics using artificial intelligence (advanced robotics)

[19] https://www.bancaditalia.it/pubblicazioni/metodi-e-fonti-note/metodi-note-2017/en_survey_methodology_invind.pdf?language_id=1

5) 3D printing

In 2017, firms were asked if they use or intend to adopt the following advanced technologies:

1) Cloud computing
2) E-commerce
3) Big data (e.g. the collection and use of large quantities of data which, also through machine learning algorithms, can assist decision-making; possible applications: distance diagnosis, financial trading algorithms, patent and legal research)
4) Internet of things (e.g. the use of technologies that, by means of advanced sensors,enable communication between the various devices used in production and business processes, facilitating their integration)
5) Artificial intelligence
6) Industrial robotics using artificial intelligence (advanced robotics)
7) 3D printing

Appendix D

Panel Analysis: Attrition

There is some attrition in the panel, as firms leave and are replaced. "The firms observed in the previous edition of the survey are always contacted again if they are still part of the target population, while those no longer wishing to take part are replaced with others in the same branch of activity and size class."[20] The main reasons given for leaving the survey are change of activity and staff cutbacks to below the entry threshold.

Year	**% in subsequent year**	**% in preceding year**
2012	0.82	-
2013	0.84	0.82
2014	0.86	0.83
2015	0.84	0.84
2016	0.86	0.87
2017	0.85	0.83
2018	-	0.88

Table D1—: *The second column contains the proportion of firms appearing in that year which are present in the panel the following year. The third contains the proportion of firms appearing in that year which were present in the panel the previous year.*

Panel Analysis: Firms Kept and Dropped

This section deals with selection issues in the panel, arising from just following firms that answered an automation technology question in 2015 and appeared in at least 3 years between 2012 - 2018. Clearly this leads to dropping firms from the full survey sample in each year, and dropping firms that answered the automation technology question that don't meet the 3 year requirement or feature in a given year.

[20] https://www.bancaditalia.it/pubblicazioni/metodi-e-fonti-note/metodi-note-2017/en_survey_methodology_invind.pdf?language_id=1

Year	% of panel	% of 2015 firms that answered dropped
2012	0.60	0.26
2013	0.64	0.21
2014	0.68	0.14
2015	0.77	0.00
2016	0.69	0.14
2017	0.63	0.18
2018	0.61	0.24

Table D2—: *The second column contains proportions of each survey year sample kept in the panel, and the third contains the proportion of firms that answered the automation technology question that are dropped.*

Appendix E

Although there are clearly significant differences between firms that automate and those that do not, it may be that the sample is not representative, even though the survey itself is balanced. This is because not all of the panel is asked these automation-related questions: construction firms are excluded, and some sectors are under-represented. I compute a logit regression on the binary variable of response to the questions on technology adoption, to see which firm characteristics are related to non-response. I find consistent results across all technologies: firms in the North-East, and in sectors SS2 (textiles, clothing, and hide, leather and footwear products) and SS7 (energy and extraction) are less likely to respond to these questions. The full results are included in Table E1.

Therefore, firms that do not answer the automation-related questions will be dropped from the subsequent panel analysis. Dealing with potentially non-random non-response requires adjusting the weights, using known information about the non-responding firms.

To this end, I update the weights supplied in the survey. I adjusts these weights depending on the scope of firms considered in any piece of analysis. Following Dargatz and Hill (1996), I use adjustment classes (sector $\times$ size) and inflate the weights of all respondents (within an adjustment class) by the class non-response adjustment factor:[21]

$$CAF_h = \frac{\sum_i w_{i,h,eligibles}}{\sum_i w_{i,h,respondents}}$$

where $w_{i,h,x}$ is the weight of firm i in adjustment class/strata h and $x \in$ [eligibles,respondents] denotes if the firm was eligible to respond to the survey, or if they actually responded. Then the non-response adjusted weight of each firm is the original weight inflated by the relevant class non-response adjustment factor: $\widetilde{w_{i,h}} = w_{i,h} \times CAF_h$.

Given that firm non-response varies systematically with sectors (see Table E1 below), I am confident that this class non-response adjustment factor should account for the selection in the survey.

Responding to the technology adoption question is regressed on industry, region, size and age of firms. The following results include just the industry and region results - the only significant results.

[21] I utilise the strata in the survey for this exercise, of which there are 66 combinations of 2-digit sector and firm size.

Table E1—: Logit Regression: Answering Technology Adoption Questions 2015

	Mobile & Cloud	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
SS2	-0.719^{***}	-0.742^{***}	-0.744^{***}	-0.643^{**}	-0.721^{***}
SS3	0.022	-0.135	-0.062	0.087	0.0215
SS4	-0.422	-0.381	-0.403	-0.290	-0.336
SS5	-0.201	-0.182	-0.168	-0.141	-0.111
SS6	-0.432^{*}	-0.455^{*}	-0.442^{*}	-0.367	-0.414
SS7	-0.960^{***}	-0.879^{***}	-0.906^{***}	-0.842^{***}	-0.873^{***}
SS8	-0.285	-0.267	-0.270	-0.218	-0.260
SS9	-0.299	-0.241	-0.254	-0.198	-0.236
SS10	-0.366	-0.341	-0.403	-0.307	-0.365
SS11	0.003	0.069	0.035	0.088	0.060
North-East	-0.506^{***}	-0.472^{***}	-0.429^{***}	-0.522^{***}	-0.495^{***}
Centre	0.139	0.173	0.183	0.114	0.139
South & Islands	0.276	0.225	0.233	0.209	0.216

Note: Estimated coefficients from logit model, regressing the response/non-response to the technology adoption question in 2015 on industry, region, size and age of firms. Estimates are significant at levels of 1%: ***, 5%: **, 10% *. Estimates for industry are with respect to the omitted industry SS1, and for region with respect to the omitted region North-West.

Appendix F

Sector	Description	ISTAT ATECO 2007
SS1	Food industries, beverages & tobacco products	10, 11, 12
SS2	Textiles, clothing, & hide, leather & footwear products	13, 14, 15
SS3	Coke manufacturing, chemical industry, rubber & plastics	19 - 22
SS4	Processing of non-metallic minerals	23
SS5	Metal engineering industry	24 - 30, 33
SS6	Other manufacturing industries	16, 17, 18, 31, 32
SS7	Other industries excl. construction (Energy & Extraction)	05 - 09, 35 - 39
SS8	Wholesale & retail commerce	45, 46, 47
SS9	Hotels & restaurants	55, 56
SS10	Transport & communications	49 - 53, 58 - 63
SS11	Real estate activities, IT, etc.	68 - 75, 77 - 82

Table F1—: *The mapping between the sectors in Bank of Italy's "Survey of Industrial and Service Firms" and the ISTAT ATECO 2007 industry classification.*

	Mobile & Cloud	AI & Big Data	IoT	Robots	3D Printing
Yes	2,524	440	642	485	360
No	934	2,953	2,742	2,912	3,037
NA	937	1,002	1,011	998	998
% Yes (all)	*0.57*	*0.10*	*0.15*	*0.11*	*0.08*
% Yes (answered)	*0.73*	*0.13*	*0.19*	*0.14*	*0.11*

Table F2—: *Answers from firms on use of advanced technologies in 2015, along with proportion of firms adopting each technology (of all firms, and of firms that were asked).*

	Cloud	**AI**	**Big Data**	**IoT**	**Robots**	**3D Printing**	**E-Comm.**
Yes	849	226	416	733	477	375	871
No	2,976	3,591	3,404	3,092	3,349	3,443	2,974
NA	566	574	571	566	565	573	546
% Yes (all)	*0.19*	*0.05*	*0.10*	*0.17*	*0.11*	*0.09*	*0.20*
% Yes (answered)	*0.22*	*0.06*	*0.11*	*0.19*	*0.12*	*0.10*	*0.23*

Table F3—: *Answers from firms on use of advanced technologies in 2017, along with proportion of firms adopting each technology (of all firms, and of firms that were asked).*

	Mobile & Cloud	**AI & Big Data**	**IoT**	**Robots**	**3D Printing**
All firms	0.56	0.09	0.12	0.06	0.05
Answered firms	0.73	0.11	0.16	0.08	0.07

Table F4—: *The weighted proportions of firms that use advanced technologies in 2015.*

	Any Tech.	**Digital Tech.**	**Physical Tech.**
All firms	0.23	0.16	0.10
Answered firms	0.30	0.21	0.13

Table F5—: *The weighted proportions of firms that use combinations of advanced technologies in 2015.*

	Cloud	**AI**	**Big Data**	**IoT**	**Robots**	**3D Printing**	**E-Comm.**
All firms	0.18	0.03	0.06	0.11	0.07	0.05	0.19
Answered firms	0.21	0.03	0.07	0.13	0.08	0.06	0.22

Table F6—: *The weighted proportions of firms that use advanced technologies in 2017.*

	Any Tech.	**Digital Tech.**	**Physical Tech.**
All firms	0.28	0.25	0.10
Answered firms	0.33	0.29	0.11

Table F7—: *The weighted proportions of firms that use combinations of advanced technologies in 2017.*

	Mobile & Cloud	AI & Big Data	IoT	Robots	3D Printing
Mobile & Cloud	1.00	0.39	0.44	0.37	0.35
AI & Big Data	-	1.00	0.83	0.81	0.82
IoT	-	-	1.00	0.80	0.79
Robots	-	-	-	1.00	0.84
3D Printing	-	-	-	-	1.00

Table F8—: *Proportion of technology adopters adopting another technology in 2015.*

	Cloud	AI	Big Data	IoT	Robots	3D Printing	E-Comm.
Cloud	1.00	0.79	0.81	0.78	0.76	0.77	0.76
AI	-	1.00	0.90	0.83	0.88	0.88	0.77
Big Data	-	-	1.00	0.84	0.84	0.85	0.77
IoT	-	-	-	1.00	0.81	0.81	0.72
Robots	-	-	-	-	1.00	0.86	0.72
3D Printing	-	-	-	-	-	1.00	0.75
E-Comm.	-	-	-	-	-	-	1.00

Table F9—: *Proportion of technology adopters adopting another technology in 2017.*

Appendix G

Propensity Score Matching - Technology Adoption

This section contains details on the Propensity Score Matching (PSM) approach used in Section III. For firms using each technology - along with those adopting 'any' technology - I run a logit on a wide set of covariates for 2015 technological adoption. The regressors are: age, turnover, investment, exporting behaviour, wage, hires per worker, fires per worker, blue-collar proportion of workers, alongside sector and region fixed-effects. Various measures of accuracy of this regression are contained below in Table G1. It is clear that the propensity scores perform reasonably well across most measures, except for precision, which is the proportion of firms predicted to adopt which actually do so. This is likely due to the relatively small share of firms which do use the technologies, hence the F1 Score is preferred to account for the unbalanced data.

Table G1—: Accuracy of Propensity Scores Predicting 2015 Technology Adoption

	Dependent variable: *2015 Tech. Adoption*					
	Any Tech.	*Cloud Computing*	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
Accuracy	0.61		0.57	0.56	0.59	0.58
Sensitivity	0.65		0.76	0.65	0.83	0.89
Specificity	0.59		0.54	0.54	0.55	0.54
Precision	0.49		0.20	0.27	0.23	0.18
Recall	0.94		0.86	0.95	0.98	0.74
F1 Score	0.59		0.33	0.41	0.38	0.31
N	2547		2541	2544	2538	2554

I also use the propensity scores to match adopters and non-adopters, and run regressions on how employment varies with technological adoption. The following table contains the number of firms considered for the matching algorithm, and the number matched. Although the sample size is lower than for other analysis in this paper (such as the panel models), the matching algorithm should reduce selection bias.

Finally, I use propensity scores from 2015 to predict adoption in 2017. The estimated coefficients are large, positive and statistically significant. These are contained in Table G3. The accuracy of these models are summarised in Tables G4 and G5.

Table G2—: Number of Matched Firms for PSM

	Any Tech.	*Cloud*	*AI & Big Data*	*IoT*	*Industrial Robotics*	*3D Printing*
Non-Adopters	1597		2210	2020	2184	2276
Adopters	957		337	521	360	262
Matched	1914		674	1042	720	524

Table G3—: Predicting 2017 Technology Adoption: Estimated Logit Coefficients

	Dependent variable: *2017 Tech. Adoption*					
	Any Tech.	*Cloud Computing*	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
Propensity Score (PS)	5.25***		5.36***	5.95***	8.08***	9.07***
	(0.38)		(0.60)	(0.76)	(0.68)	(0.68)
$\mathbb{1}$(PS > Median)	1.17***		0.87***	0.92***	1.76***	2.32***
	(0.09)		(0.14)	(0.12)	(0.17)	(0.24)
N	2003		1996	1994	1998	1988

Note: Estimates are significant at levels of 0.1%: ***, 1%: **, 5% *.

Table G4—: Predicting 2017 Technology Adoption: F1 Score

	Dependent variable: *2017 Tech. Adoption*					
	Any Tech.	*Cloud Computing*	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
Propensity Score (PS)	0.62		0.30	0.43	0.37	0.34
$\mathbb{1}$(PS > Median)	0.63		0.32	0.42	0.36	0.30
2017 Variables	0.65		0.35	0.48	0.40	0.36

Note: The F1 Scores contained here are the harmonic mean of Precision (fraction of firms predicted as technology adopters that actually are) and Recall (fraction of technology adopters correctly predicted).

Table G5—: Predicting 2017 Technology Adoption: Accuracy

	Dependent variable: *2017 Tech. Adoption*					
	Any Tech.	*Cloud Computing*	*AI & Big Data*	*Internet of Things*	*Industrial Robotics*	*3D Printing*
Propensity Score (PS)	0.62		0.44	0.59	0.64	0.65
$\mathbb{1}$(PS > Median)	0.64		0.55	0.57	0.59	0.58
2017 Variables	0.62		0.40	0.55	0.59	0.60

Note: The Accuracy measure is the number of true positives and true negatives, divided by the full sample.

Appendix H

In Section III, I presented results from TWFE models on the relationship between adoption of automation technologies and firm outcomes: overall employment, productivity and blue-collar employment.

However, there are potential threats to identification in these specifications. It is plausible that an omitted variable is correlated with both technology adoption and the various dependent variables. For example, a change in management could lead to a new strategy of aggressive growth and adoption of new technologies. Therefore, I instrument technology adoption with various sets of plausibly valid and exogenous instruments. I leverage the full panel to use a time series of instruments from 2010 to 2018.[22] The instruments used are intangible goods expenditure, expenditure on R&D, and fixed investment expenditure. The identifying assumption is that each of these series is correlated with technological adoption decisions in each year, but do not directly influence employment decisions or firm turnover. This seems possible for the former two: spending on intangible goods and R&D. However, it is reasonable to expect that fixed investment expenditure would not qualify here: to the extent that labour is chosen to equate its marginal product (MPL) to the real wage, and to the degree that investment affects the MPL. Nevertheless, it is possible that lagged values of fixed investment could satisfy the exclusion restriction.

Table H1—: Estimates of β from TWFE models, instrumented by expenditure on R&D

Technology	*Dependent Variable*	*Coefficient*	*F-stat.*	*Cond. F-stat.*	*Hausman*	*Sargan*
Cloud Computing	Employment	0.836^{**}	2.143	0.838	4.709	2.767
	Blue-collar Emp.	-0.647	1.691	0.507	0.817	0.390
	Turnover per worker	-0.036	2.143	0.838	0.008	9.413
AI & Big Data	Employment	0.609^{*}	4.047	1.584	2.685	8.660
	Blue-collar Emp.	-0.306	5.594	1.678	0.451	0.640
	Turnover per worker	-0.01	4.047	1.584	0.000	9.558
IoT	Employment	0.412^{**}	6.108	2.39	3.378	2.390
	Blue-collar Emp.	0.169	7.825	2.347	0.205	0.568
	Turnover per worker	0.008	6.108	2.39	0.002	9.618
Industrial Robotics	Employment	0.516^{**}	5.227	2.045	3.972	10.70
	Blue-collar Emp.	-0.02	6.829	2.049	0.125	0.575
	Turnover per worker	-0.064	5.227	2.045	0.237	9.684
3D Printing	Employment	0.478^{**}	5.869	2.296	3.646	15.60
	Blue-collar Emp.	0.147	8.027	2.408	0.664	0.655
	Turnover per worker	0.033	5.869	2.296	0.005	10.42

Notes: Robust standard errors clustered at firm level. Coefficients labelled by statistical significance at: *** 0.1%, ** 1%, * 5%. The test statistics are: first-stage F-statistic; conditional F-statistic (Sanderson and Windmeijer, 2016); Hasuman Test for endogeneity; Sargan statistic for over-identifying restrictions.

The results for the IV regressions are presented in Table H1, with the R&D expenditure

[22] They only cover 2010 - 2015 for blue-collar employment, due to data limitations.

instrument. It contains the estimated coefficients and associated tests for each technology, for the three dependent variables. I find significant, and much larger estimated coefficients for the impact of technological adoption on employment, across all technologies. Although there is evidence of OLS endogeneity in these specifications (large Hausman values), and instrument validity (large Sargan statistics), it is likely the results suffer somewhat from a weak instrument problem (F-statistics are lower than typical critical values (Stock and Yogo, 2005)).

The estimates for unskilled employment and turnover per worker are all statistically insignificant in the IV setting, while evidence of OLS consistency, and invalid and weak instruments, is quite substantial.

I take this evidence to suggest that there is some endogeneity in my estimates for the impact of technological adoption on total employment, which bias the initial estimates towards zero. However, due to the likelihood my instruments are weak, these IV estimates should be treated with caution.

Appendix I

As a starting point to the value function iteration, I give the algorithm the static optimal input choices. These are computed below.

$$\pi_t(z_t, n_{t-1}) = \max_{n_t^r, n_t^n \geq 0} \{p_t z_t (n_t^n)^\alpha (n_t^r)^\gamma - w_t^n n_t^n - w_t^n n_t^r - g(n_t, n_{t-1}) - c_f\}$$
$$[n_t^r] : \gamma p_t z_t (n_t^n)^\alpha (n_t^r)^{\gamma-1} - w_t^n - \tau(n_t - n_{t-1}) = 0$$
$$[n_t^n] : \alpha p_t z_t (n_t^n)^{\alpha-1} (n_t^r)^\gamma - w_t^r - \tau(n_t - n_{t-1}) = 0$$

Combining the FOCs:

$$\frac{\alpha}{\gamma}\frac{n_t^r}{n_t^n} = \frac{w_t^n}{w_t^r} + 1$$

which can be rearranged for n_t^r, n_t^n and plugged back into the production function. Thus the conditional labour demands are:

$$n_t^r = \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^\alpha\right]^{\frac{1}{\alpha+\gamma}}$$
$$n_t^n = \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}}$$

These equations are substituted back into the static profit formula, yielding a function in terms of y_t. Taking the derivative with respect to y_t yields the output the firm chooses to maximise profit.

$$\pi_t(z_t, n_{t-1}) = p_t y_t - w_t^n \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^\alpha\right]^{\frac{1}{\alpha+\gamma}} - w_t^n \left[\frac{y_t}{z_t}\left(\frac{-\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}}$$
$$- \frac{\tau}{2}\left(\left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^\alpha\right]^{\frac{1}{\alpha+\gamma}} + \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}} - n_{t-1}\right)^2 - c_f$$

$$[y_t] : p_t - \frac{1}{\alpha+\gamma} y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}} z_t^{-\frac{1}{\alpha+\gamma}} \left(w_t^n \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{\alpha}{\alpha+\gamma}} + w_t^r \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{-\gamma}{\alpha+\gamma}}\right)$$
$$- \tau\left[\left(\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{\alpha}{\alpha+\gamma}} + \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{-\gamma}{\alpha+\gamma}}\right)\right.$$
$$\left.\frac{1}{\alpha+\gamma} y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}} z_t^{-\frac{1}{\alpha+\gamma}} \left(\left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^\alpha\right]^{\frac{1}{\alpha+\gamma}} + \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}} - n_{t-1}\right)\right] = 0$$

Let $\psi = \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)$ to simplify the final condition.

$$p_t(\alpha+\gamma)z_t^{\frac{1}{\alpha+\gamma}} = y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}}\Bigg[\big(w_t^n\psi^{\frac{\alpha}{\alpha+\gamma}} + w_t^r\psi^{\frac{-\gamma}{\alpha+\gamma}}\big) + \tau\Bigg(\big(\psi^{\frac{\alpha}{\alpha+\gamma}} + \psi^{\frac{-\gamma}{\alpha+\gamma}}\big)\Big(\big(\frac{y_t}{z_t}\big)^{\frac{1}{\alpha+\gamma}}\big(\psi^{\frac{\alpha}{\alpha+\gamma}} + \psi^{\frac{-\gamma}{\alpha+\gamma}}\big) - n_{t-1}\Big)\Bigg)\Bigg]$$

Then a nonlinear solver can be used to find $y_t(z_t, n_{t-1})$ over a grid for z_t, n_{t-1} which satisfies this FOC.
For automating firms:

$$\pi_t^A(z_t, n_{t-1}) = \max_{R_t, n_t^r, n_t^n \geq 0}\{p_t z_t (n_t^n)^\alpha (R_t + n_t^r)^\gamma - w_t^n n_t^n - w_t^n n_t^r - R_t q_t - g(n_t, n_{t-1}) - c_f\}$$
$$[n_t^r] : \gamma p_t z_t (R_t + n_t^n)^\alpha (n_t^r)^{\gamma-1} - w_t^n - \tau(n_t - n_{t-1}) = 0$$
$$[n_t^n] : \alpha p_t z_t (R_t + n_t^n)^{\alpha-1} (n_t^r)^\gamma - w_t^r - \tau(n_t - n_{t-1}) = 0$$
$$[R_t] : \gamma p_t z_t (R_t + n_t^n)^\alpha (n_t^r)^{\gamma-1} - q_t = 0$$

Combing the FOCs:

$$\frac{\alpha}{\gamma}\frac{R_t + n_t^r}{n_t^n} = \frac{w_t^n}{w_t^r} + 1 \tag{I1}$$

$$1 = \frac{w_t^r}{q_t} + \tau(n_t - n_{t-1}) \tag{I2}$$

as before equation I1 is rearranged, but now for n_t^n and $R_t + n_t^r$, before plugging back into the production function. The result is conditional demands for these two objects.

$$n_t^r + R_t = \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^\alpha\right]^{\frac{1}{\alpha+\gamma}}$$
$$n_t^n = \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}}$$

To back out the demand for routine labour, and for automation capital, take I2 and rearrange to get:

$$\frac{1}{\tau}\big(1 - \frac{w_t^r}{q_t}\big) + n_{t-1} = n_t^n + n_t^r$$

Now it is easy to find conditional routine labour and automation demands:

$$n_t^r = \frac{1}{\tau}\big(1 - \frac{w_t^r}{q_t}\big) + n_{t-1} - \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}}$$

$$R_t = \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\alpha}\right]^{\frac{1}{\alpha+\gamma}} - n_t^r$$

Substituting these back into the profit function and taking the derivative with respect to y_t yields the following condition:

$$\begin{aligned}
[y_t] : p_t &- \frac{1}{\alpha+\gamma} y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}} z_t^{-\frac{1}{\alpha+\gamma}} \left(w_t^n \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{\alpha}{\alpha+\gamma}} + w_t^r \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{-\gamma}{\alpha+\gamma}} \right) \\
&- q_t \frac{1}{\alpha+\gamma} y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}} z_t^{-\frac{1}{\alpha+\gamma}} \left(\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{\alpha}{\alpha+\gamma}} + \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{-\gamma}{\alpha+\gamma}} \right) \\
&- \tau \Bigg[\left(\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{\alpha}{\alpha+\gamma}} + \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\frac{-\gamma}{\alpha+\gamma}} \right) \\
&\frac{1}{\alpha+\gamma} y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}} z_t^{-\frac{1}{\alpha+\gamma}} \left(\left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{\alpha}\right]^{\frac{1}{\alpha+\gamma}} + \left[\frac{y_t}{z_t}\left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)^{-\gamma}\right]^{\frac{1}{\alpha+\gamma}} - n_{t-1} \right) \Bigg] = 0
\end{aligned}$$

Again let $\psi = \left(\frac{\gamma}{\alpha}\frac{w_t^n + w_t^r}{w_t^r}\right)$ to simplify the final condition.

$$\begin{aligned}
p_t(\alpha+\gamma) z_t^{\frac{1}{\alpha+\gamma}} = y_t^{\frac{1-\alpha-\gamma}{\alpha+\gamma}} \Bigg[&\big(w_t^n \psi^{\frac{\alpha}{\alpha+\gamma}} + w_t^r \psi^{\frac{-\gamma}{\alpha+\gamma}}\big) + q_t\big(\psi^{\frac{\alpha}{\alpha+\gamma}} + \psi^{\frac{-\gamma}{\alpha+\gamma}}\big) + \\
&\tau\left(\big(\psi^{\frac{\alpha}{\alpha+\gamma}} + \psi^{\frac{-\gamma}{\alpha+\gamma}}\big)\Big(\big(\frac{y_t}{z_t}\big)^{\frac{1}{\alpha+\gamma}}\big(\psi^{\frac{\alpha}{\alpha+\gamma}} + \psi^{\frac{-\gamma}{\alpha+\gamma}}\big) - n_{t-1}\Big) \right) \Bigg]
\end{aligned}$$